Exploration and Safety in Model-Based Reinforcement Learning

Guest lecture, MAE 207 Safety for Autonomous Systems (UC San Diego)

Felix Berkenkamp · Research Scientist, Bosch Center for AI · March 11, 2021
Video: https://www.youtube.com/watch?v=umxcj3nvScY

THE TALK IN 5 POINTS

- **Problem:** RL normally learns by trial and error in the same world it is used in. A real robot is expensive and can break, so learning must use little data and stay safe the whole time.
- **Two kinds of uncertainty:** *aleatoric* = random noise you can never learn away; *epistemic* = not knowing the dynamics *f*, which shrinks with data. Treating the second as noise can make learning fail.
- **Exploration:** greedy (trust the model's best guess) can get stuck. **Optimism** (pick the max of mean + β·std, GP-UCB) provably converges: each step's gap to the optimum is at most 2β·σ, and σ shrinks as data comes in.
- **Model-based RL:** H-UCRL makes optimism practical by adding a fake "hallucinated" control η that can push predictions anywhere inside the model's error band. It beats greedy once actions cost something.
- **Safety:** being pessimistic about safety is easy, but bolting it onto an optimistic algorithm gets stuck inside the starting safe region. Exploring the most uncertain safe points fixes this but wastes data. Doing both well is still an open problem.

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TERMS

Model-based RL: learn a model of the dynamics *f* from data, then choose the policy using that model.

Aleatoric / epistemic: noise in the system itself / uncertainty from not knowing *f*. Only the second shrinks with data.

Gaussian process (GP): a model that gives a best guess $\mu(\theta)$ and an uncertainty $\sigma(\theta)$ for every input θ.

Well-calibrated: the true function stays inside $\mu \pm \beta\cdot\sigma$ with probability $\geq 1 - \delta$.

Bandit / Bayesian optimization: the one-step version. Pick parameters θ, get a noisy score $J(\theta)$ + noise.

Regret: $J(\theta^*) - J(\theta_t)$, the gap between the best score and the one you got.



WHY SAFE, DATA-EFFICIENT RL

Title and introduction

▶ 0:00

- The host introduces Felix Berkenkamp: PhD at ETH Zurich, one of the early researchers on **safe learning**, now a research scientist at the Bosch Center for AI.
- Goal of the lecture: link what the course covered (keeping a system safe when you have a measure of uncertainty) with the machine-learning side: **how to collect data**, and how safety limits affect learning.
- He keeps it fairly high level with some math, and asks students to interrupt with questions.



WHY SAFE, DATA-EFFICIENT RL

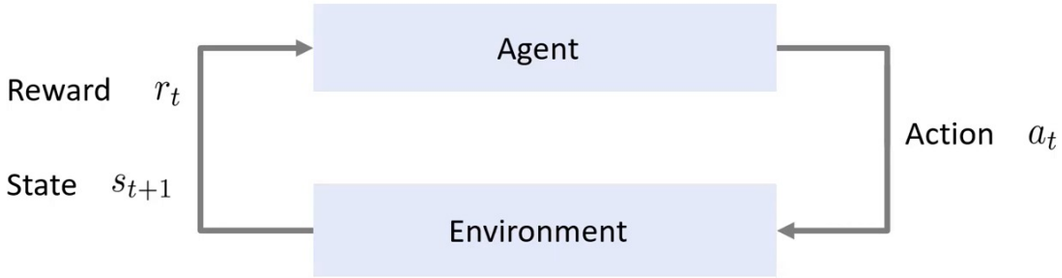
Bosch and AI

▶ 2:32

- Bosch is a traditional engineering company: more than 240 plants worldwide, deep know-how in making physical products work reliably.
- It now invests heavily in AI, because AI could make manual design work much faster, which threatens its core business if it falls behind.
- For designing controllers, **reinforcement learning (RL)** is an obvious tool to look at.

Reinforcement Learning

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Goal: Maximize sum of expected rewards

Need to interact and trade off exploration & exploitation

Reinforcement Learning: An Introduction
R. Sutton, A.G. Barto, 1998

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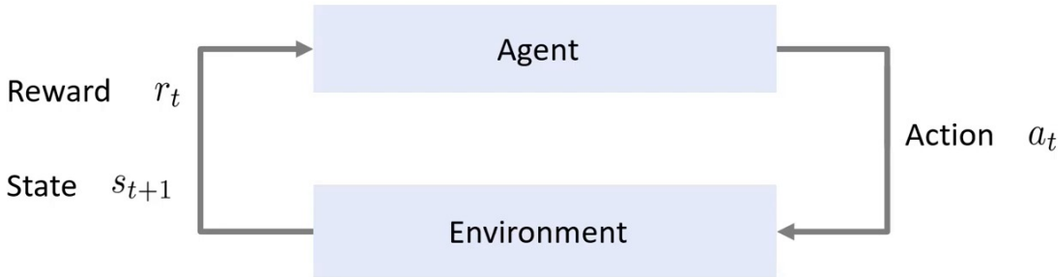
WHY SAFE, DATA-EFFICIENT RL

Reinforcement learning in one picture

▶ 3:19

- An **agent** picks an action a_t (e.g. from a control policy). The **environment** returns the next state s_{t+1} and a reward r_t saying how good that step was.
- Goal: maximize the **sum** of expected rewards over time, not just the next reward.
- The core difficulty is the **exploration–exploitation** trade-off: collect data to learn how the system reacts (explore) vs. use what you know to earn reward (exploit).

Reinforcement Learning
Assumption



Train-Environment = Test-Environment

What if
- experiments are expensive?
- the system is safety-critical?

Reinforcement Learning: An Introduction
R. Sutton, A.G. Barto, 1998

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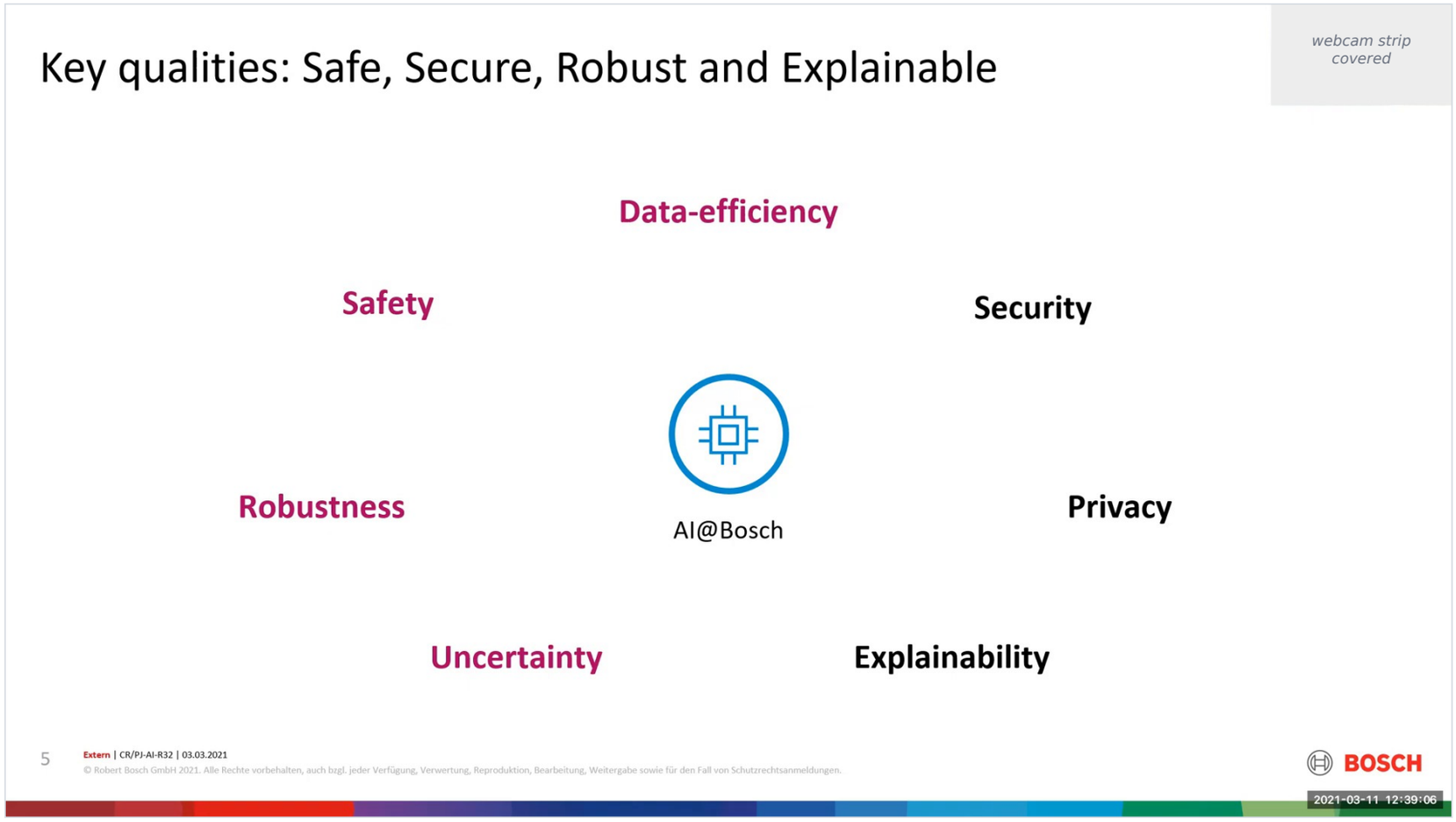
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WHY SAFE, DATA-EFFICIENT RL

RL's hidden assumption

▶ 4:46

- RL assumes the **training environment is the test environment**. There is no separate test set as in supervised learning: RL in effect overfits to the world it trains in.
- That is fine for board games or video games you can replay forever.
- It breaks when experiments are **expensive** (a physical robot) or **safety-critical**. A robot that crashes into walls while exploring randomly may destroy itself, and all learning progress is lost.

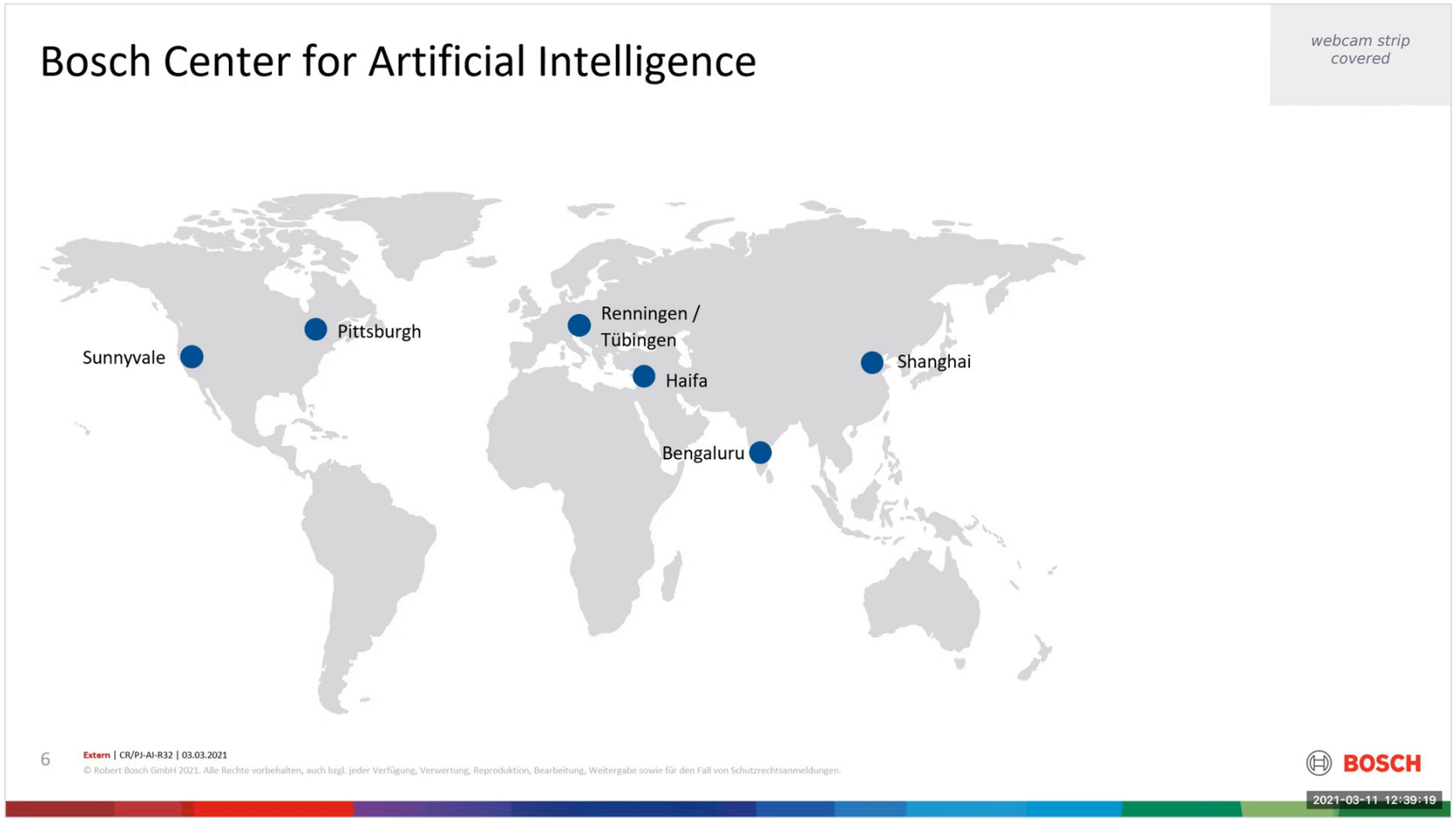


WHY SAFE, DATA-EFFICIENT RL

What Bosch cares about in AI

▶ 6:11

- Bosch's list of key qualities: data-efficiency, safety, security, privacy, robustness, explainability and uncertainty.
- Big tech mostly works on data-rich software. Bosch focuses on **physical systems**, where data is costly and mistakes are dangerous.
- This lecture covers the highlighted half: **data-efficient** learning, **safety** during learning, **robustness**, and the role of **uncertainty** in all three.

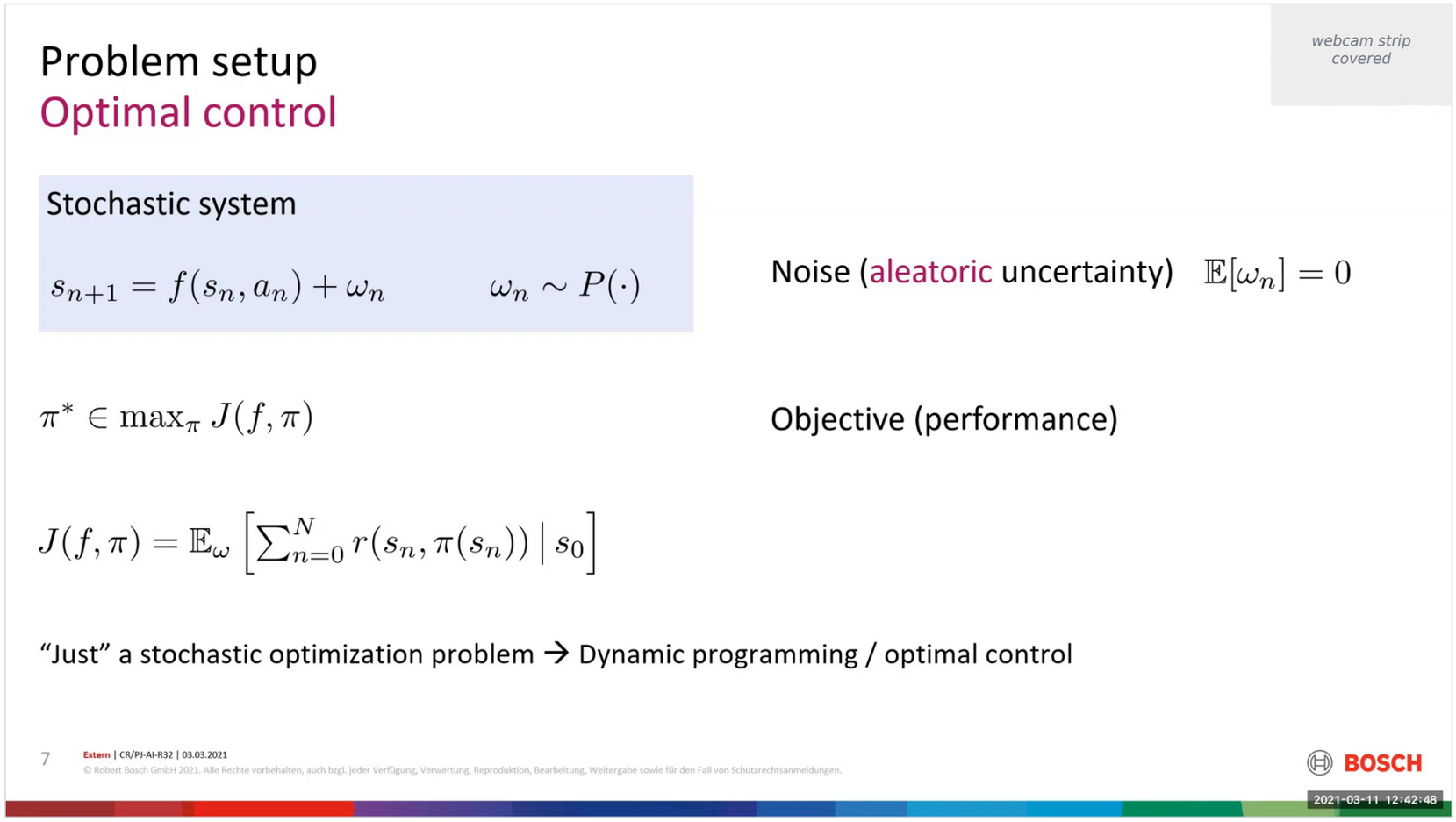


WHY SAFE, DATA-EFFICIENT RL

Bosch Center for AI

▶ 6:51

- The lab is spread over Sunnyvale, Pittsburgh, Renningen/Tübingen, Haifa, Bengaluru and Shanghai.
- A short recruiting pitch: consider Bosch for a job after your studies.

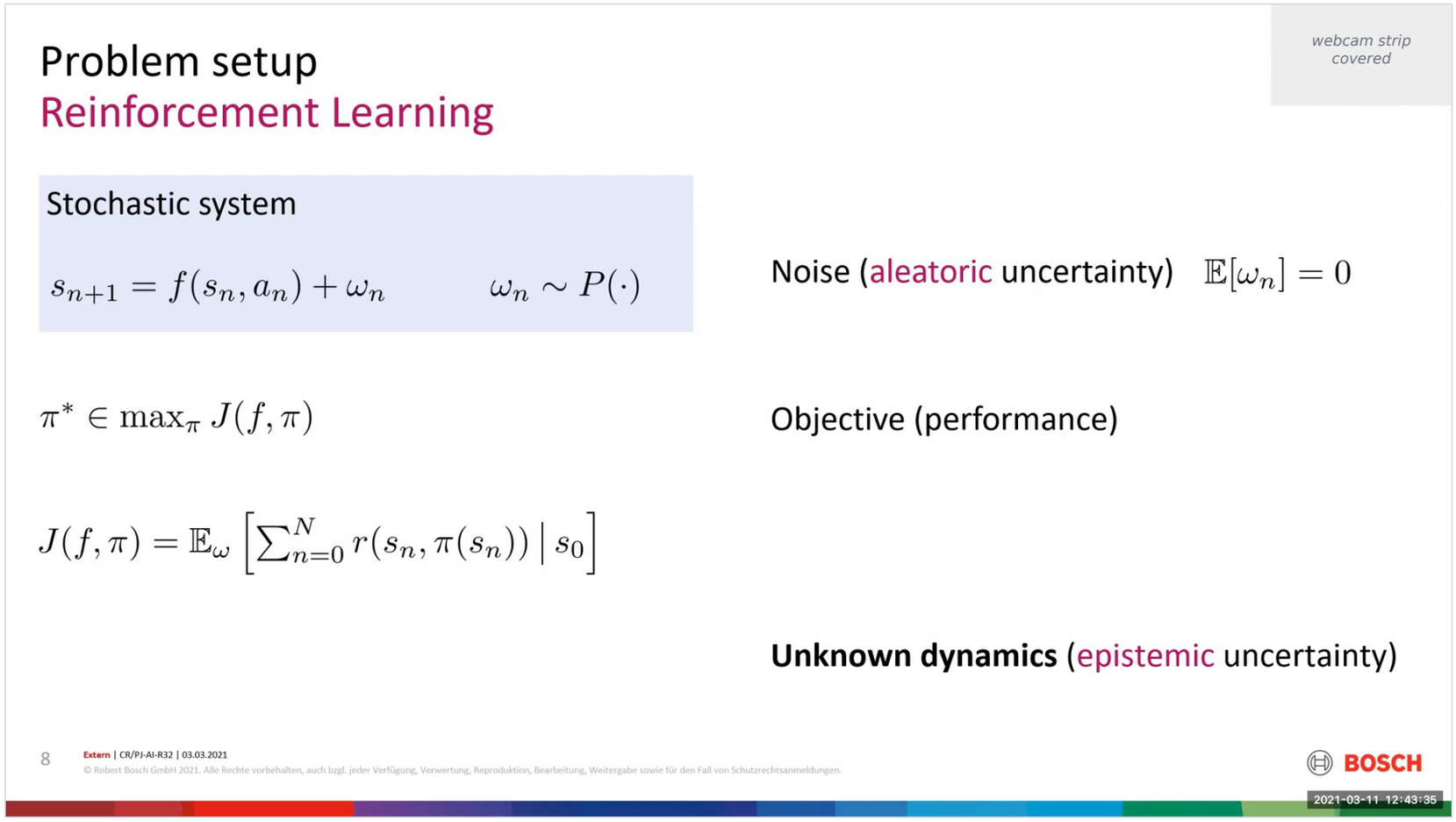


PROBLEM SETUP

Problem setup: optimal control

▶ 7:04

- State evolves as $s_{n+1} = f(s_n, a_n) + \omega_n$. *f* is a fixed (deterministic) dynamics function; ω_n is random noise drawn fresh each step. The state is fully observed.
- Assume the noise has mean 0, so many repeats from the same state and action average out to f. This noise is **aleatoric uncertainty**: every run differs.
- Objective: J(f, π) = expected sum of rewards over N steps from start s_0, averaged over the noise. Find the policy π* that maximizes it.
- If *f* is known this is “just” stochastic optimization, solved by dynamic programming / optimal control. (He writes the state as s, the ML habit, instead of x.)



PROBLEM SETUP

Problem setup: RL means f is unknown

▶ 10:33

- Same problem, but the dynamics *f* are **unknown**. This is **epistemic uncertainty**: the world follows fixed physical laws, we just don't know them.
- You could hand-build a perfect model for each new system, but that costs time and money. The hope of RL is to skip that step and learn directly.

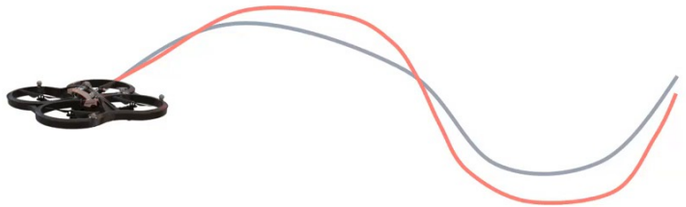
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Problem setup

Epistemic v.s. aleatoric uncertainty

$$s_{n+1} = f(s_n, a_n) + \omega_n$$

Aleatoric: Unlearnable (stochastic) randomness ω_n



We average over this as part of $J(f, \pi)$

Epistemic: Uncertainty about deterministic function f

Not part of $J(f, \pi)$

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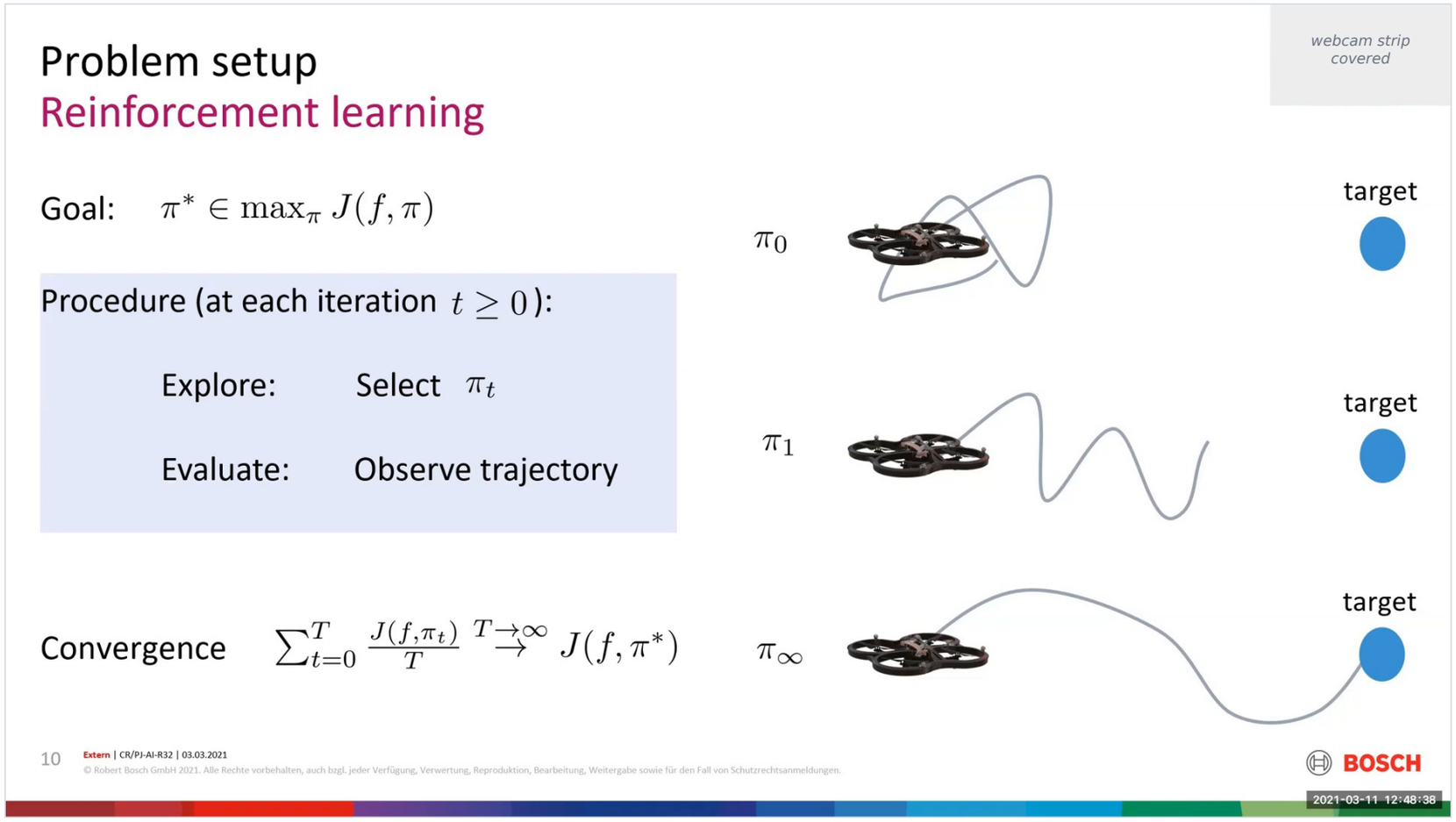
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PROBLEM SETUP

Epistemic vs. aleatoric uncertainty

▶ 11:22

- **Aleatoric**: randomness ω_n you can never learn away. Fly the quadrotor with the same policy from the same start twice and you get two different paths (grey vs. red). J averages over it.
- **Epistemic**: not knowing *f*, e.g. the drone's mass, size or exact aerodynamics. It is *not* part of the objective; we only care about the real system.
- Key difference: enough data removes epistemic uncertainty, but never aleatoric.

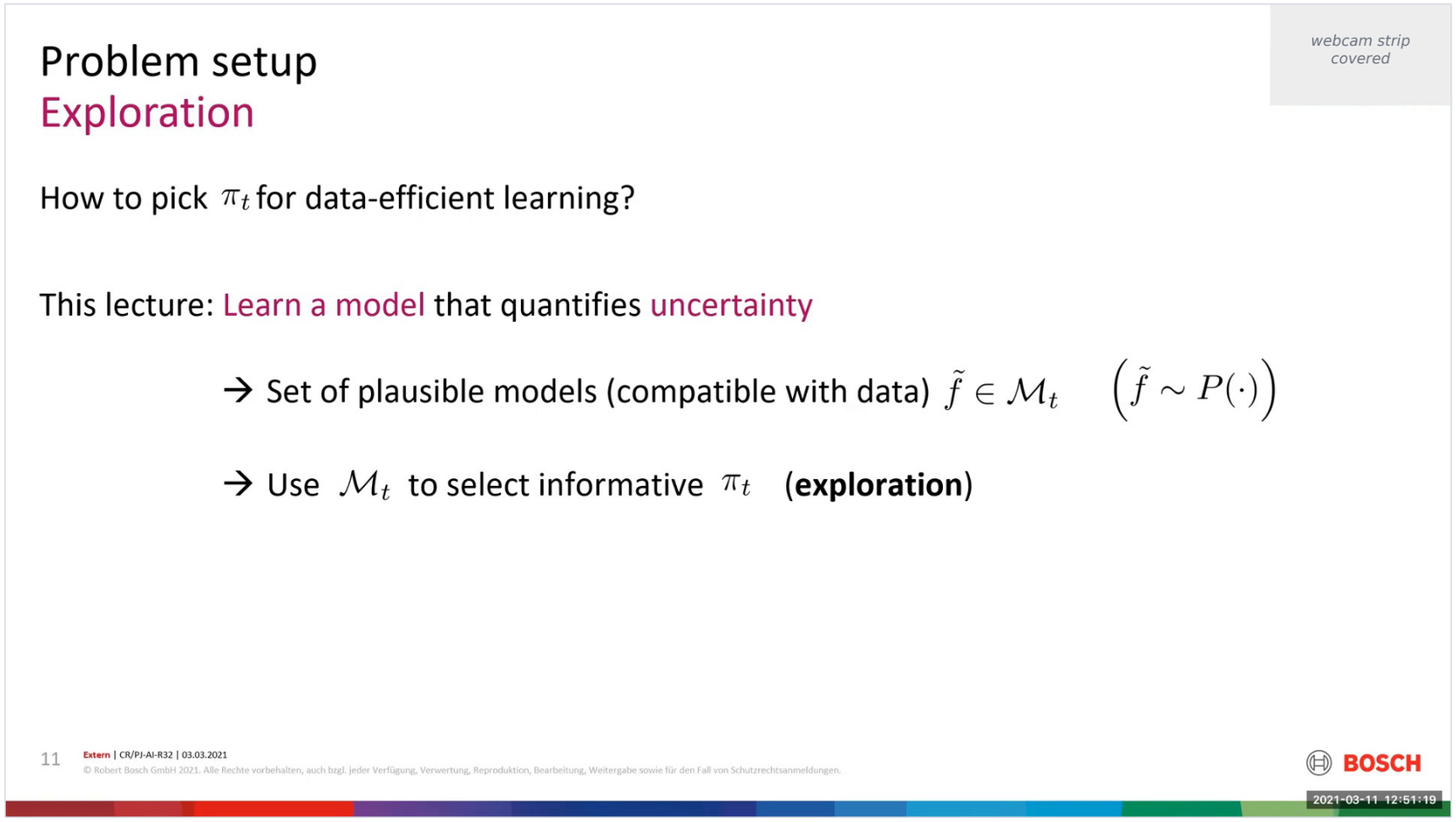


PROBLEM SETUP

The RL loop and what “converge” means

▶ 12:28

- Since *f* is unknown, we must interact. Each iteration t: pick a policy π_t (e.g. PID gains or neural-net weights), run it, record the trajectory.
- Example: π_0 wanders randomly; π_1, chosen using that data, does better; ideally π_∞ flies straight to the target.
- Convergence goal: the **average** score of all policies tried, $(1/T)\cdot\Sigma\, J(f, \pi_t)$, tends to the best score $J(f, \pi^*)$ as $T \to \infty$.
- Each score is at most the optimum, so the average can only reach it if almost all later policies are optimal. Occasional bad exploratory tries are still allowed.

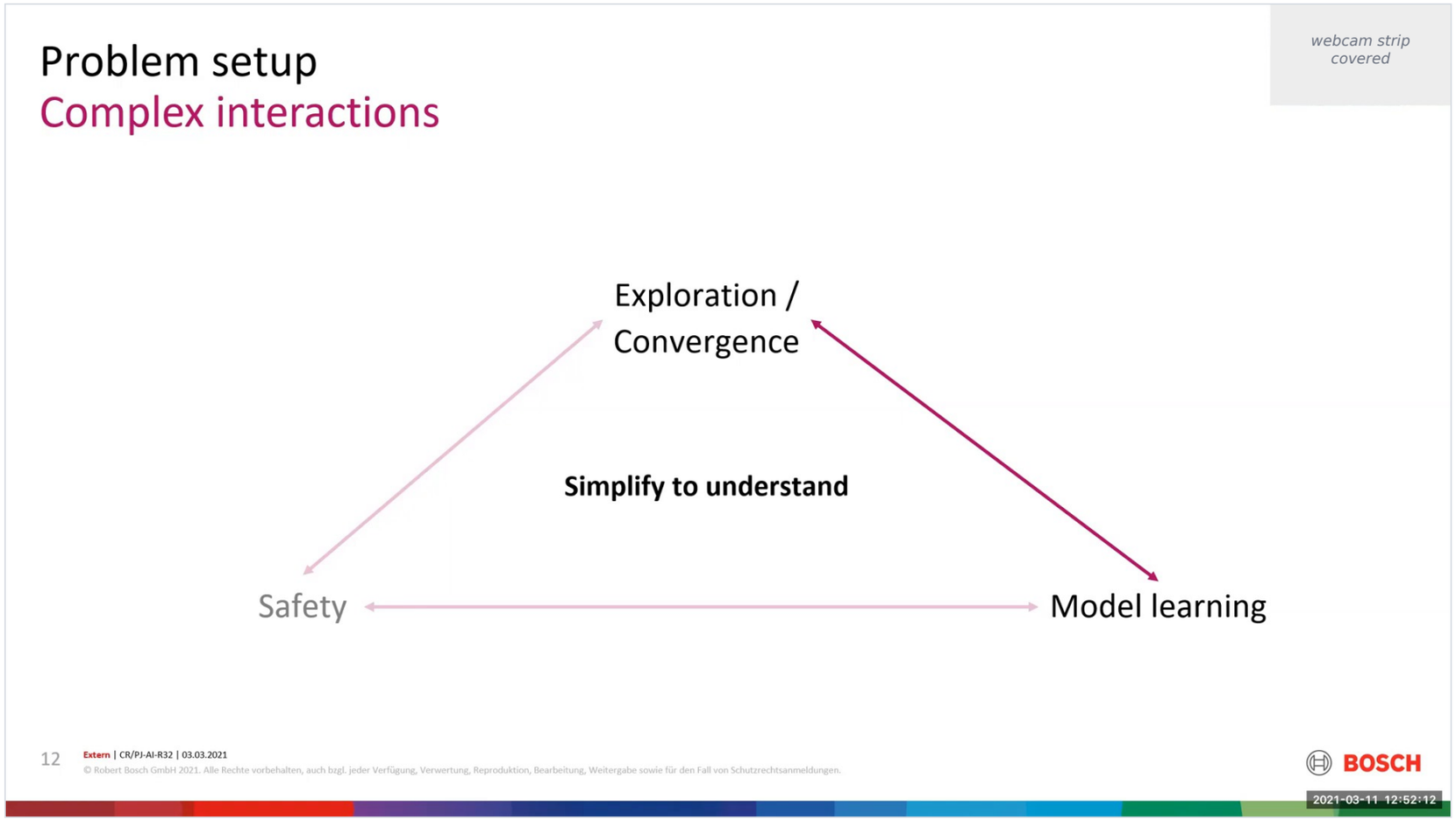


PROBLEM SETUP

How to pick the next policy: learn a model

▶ 16:23

- Choosing π_t well is the hard part. One option is model-free gradient methods; this lecture does **model-based RL** instead.
- Keep a **set of plausible models** M_t: every dynamics function $\tilde{f}$ still consistent with the data. A probabilistic belief $\tilde{f} \sim P(\cdot)$ can be turned into such a set using confidence intervals.
- Then use M_t to choose a π_t that is **informative**, i.e. teaches us about the optimum. That is exploration.

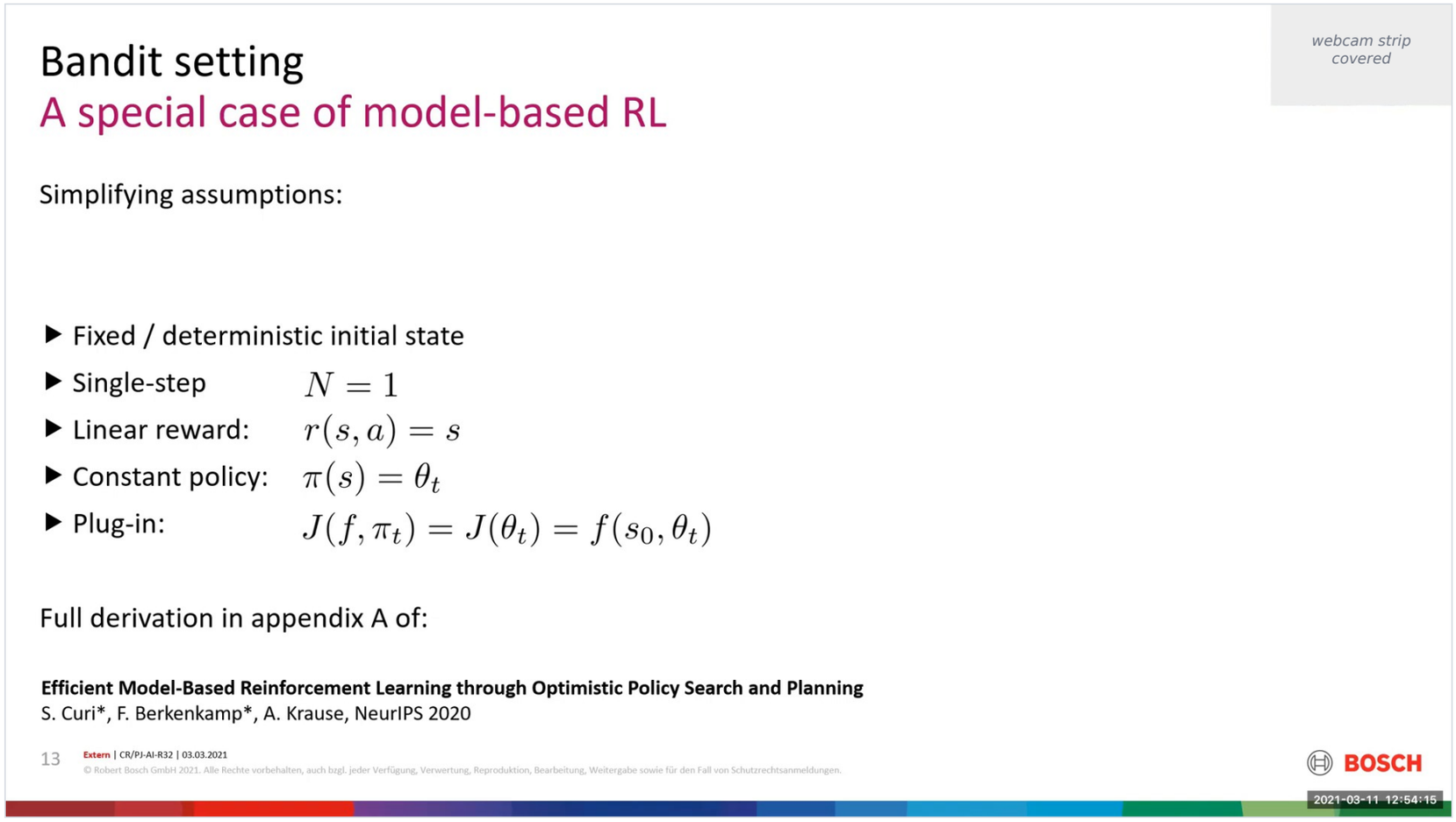


PROBLEM SETUP

Three things that affect each other

▶ 19:04

- Exploration/convergence, model learning and (later) safety all influence each other.
- How you learn the model changes how you explore. How you explore decides what data the model gets.
- Safety will make this even harder, so first simplify the problem to understand these links.

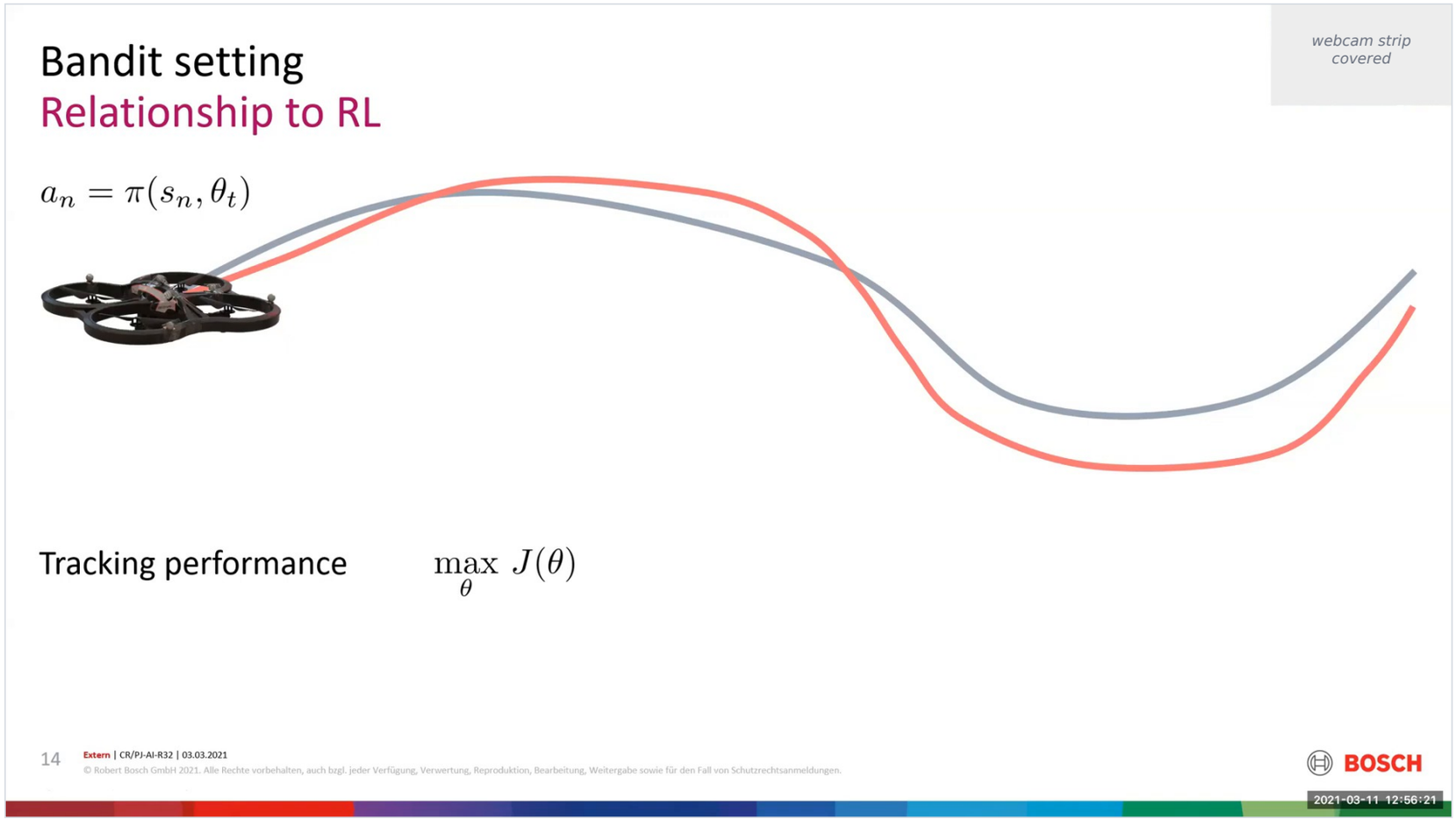


BANDITS: EXPLORING WITH UNCERTAINTY

Bandits: a special case of model-based RL

▶ 19:57

- Simplifying assumptions: fixed start state; one step only (N = 1); reward = the next state, r(s, a) = s; policy is a constant parameter vector, $\pi(s) = \theta_t$.
- Plug these in and the messy objective collapses to $J(f, \pi_t) = J(\theta_t) = f(s_0, \theta_t)$: an unknown function of θ that we want to maximize.
- The assumptions are silly in practice, but the problem keeps the key difficulty of model-based RL. Full derivation: appendix A of Curi, Berkenkamp, Krause, NeurIPS 2020.



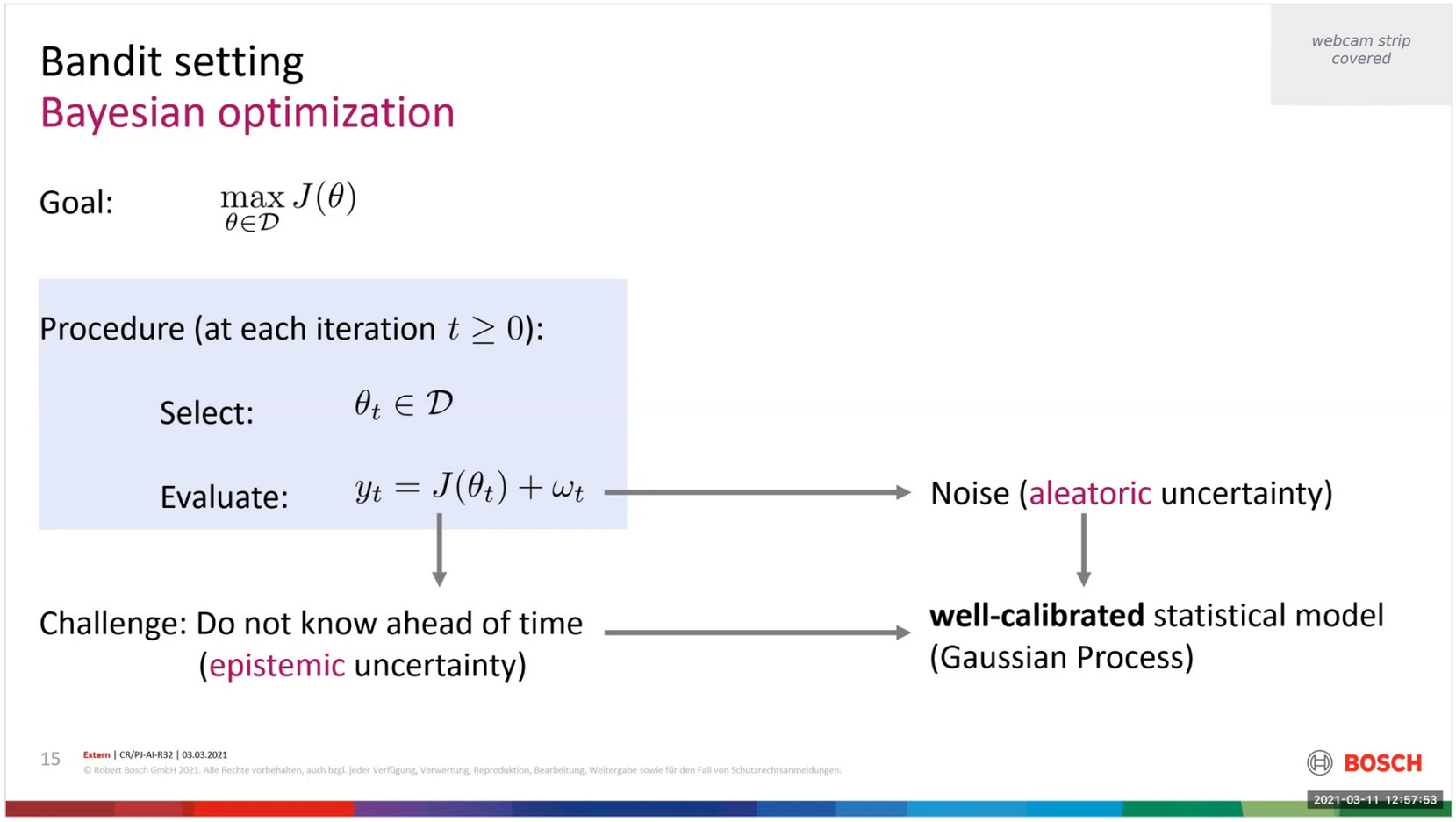
BANDITS: EXPLORING WITH UNCERTAINTY

Bandits in robot terms

▶ 22:00

- Another view: a fixed policy $a_n = \pi(s_n, \theta_t)$ with tunable parameters θ, e.g. PD gains or network weights.
- Try one θ, fly one trajectory, measure how well it tracked the path. That gives a **noisy sample** of J(θ). Goal: $\max_\theta J(\theta)$.
- No dynamics model is needed here; we model the score J(θ) directly.

Q&A: *"Maximize tracking error?" A slip. RL maximizes reward, so J is the negative tracking error (equivalently, minimize the error).*



BANDITS: EXPLORING WITH UNCERTAINTY

Bayesian optimization

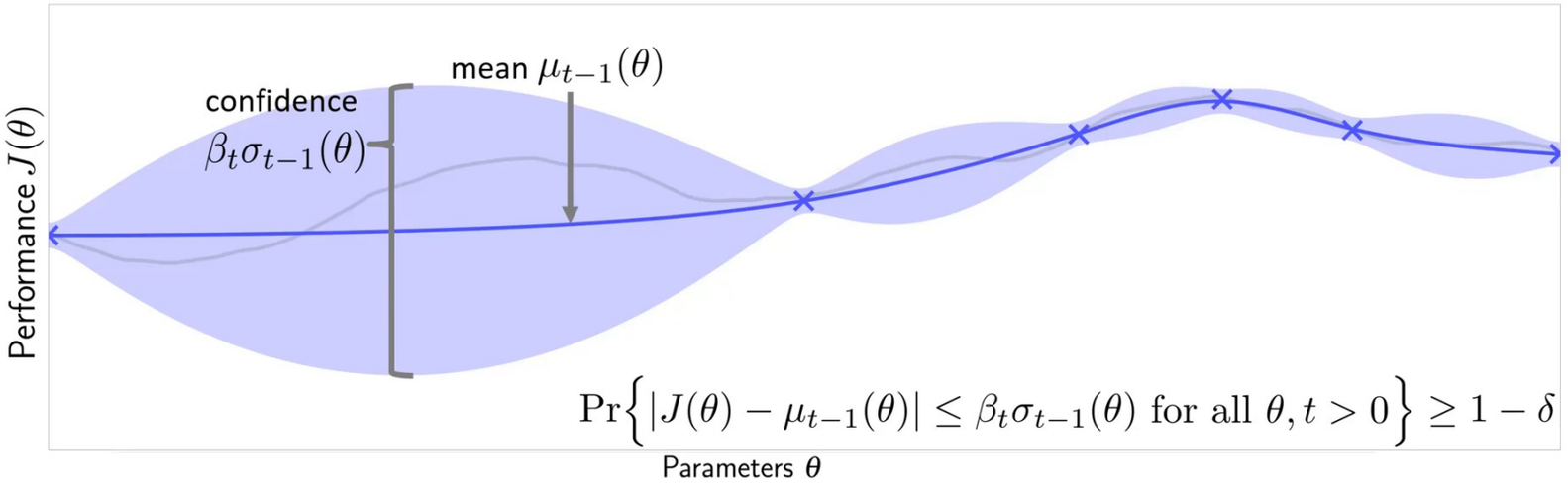
▶ 24:06

- Goal: max over θ in a set D of J(θ). Each iteration: pick θ_t, observe $y_t = J(\theta_t) + \omega_t$.
- Two uncertainties again: the noise ω_t is aleatoric; not knowing J is epistemic.
- Plan: learn a **well-calibrated** statistical model of J, here a **Gaussian process** (GP), and use it to choose θ.

Bandit setting
Well-calibrated model

webcam strip covered

well-calibrated = “Confidence intervals always contain the true function with high probability”



Information-Theoretic Regret Bounds for Gaussian Process Optimization in the Bandit Setting
N. Srinivas, A. Krause, S. Kakade, M. Seeger, IEEE Transactions on Information Theory 2012

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BANDITS: EXPLORING WITH UNCERTAINTY

Well-calibrated model

▶ 25:38 · revisited at 27:29, 28:04

- Plot: x-axis = parameters θ, y-axis = performance. Grey = true J (unknown). Crosses = measurements. Blue line = GP mean $\mu_{t-1}(\theta)$, the best guess. Shaded band = confidence $\beta_t \cdot \sigma_{t-1}(\theta)$.
- Uncertainty is small near data and grows with distance from it. Each new point shrinks the band.
- **Well-calibrated** means the band holds the truth: $\Pr\{ |J(\theta) - \mu_{t-1}(\theta)| \leq \beta_t \sigma_{t-1}(\theta) \text{ for all } \theta, t \} \geq 1 - \delta$.
- This is a strong assumption on real robots, but nothing works without it. For GPs it can be proven for suitable function classes (Srinivas et al., 2012).

Q&A: *Why is it so strong? You must assume something about J, e.g. that it changes smoothly with θ. A jumpy, discontinuous system can’t be learned. GPs suit smooth, low-dimensional functions; neural networks can also give uncertainty but are harder to interpret.*

Bandit setting
Bayesian optimization

webcam strip covered

Goal: $\max_{\theta \in \mathcal{D}} J(\theta)$

Procedure (at each iteration $t \geq 0$):

Select: $\theta_t \in \mathcal{D}$

Evaluate: $y_t = J(\theta_t) + \omega_t$

Uncertainty over $J(\theta)$

How should we select $\theta_t \in \mathcal{D}$?

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BANDITS: EXPLORING WITH UNCERTAINTY

Back to the loop: how to choose θ?

▶ 33:27

- We now have a model with uncertainty over J(θ).
- Same loop as before: select $\theta_t \in D$, evaluate $y_t = J(\theta_t) + \omega_t$.
- Open question: **how should we pick θ_t**?

Model-based RL
Exploration schemes

webcam strip covered

Goal: $\max_{\theta \in \mathcal{D}} J(\theta)$

Greedy $\theta_t = \max_\theta \mu_{t-1}(\theta)$

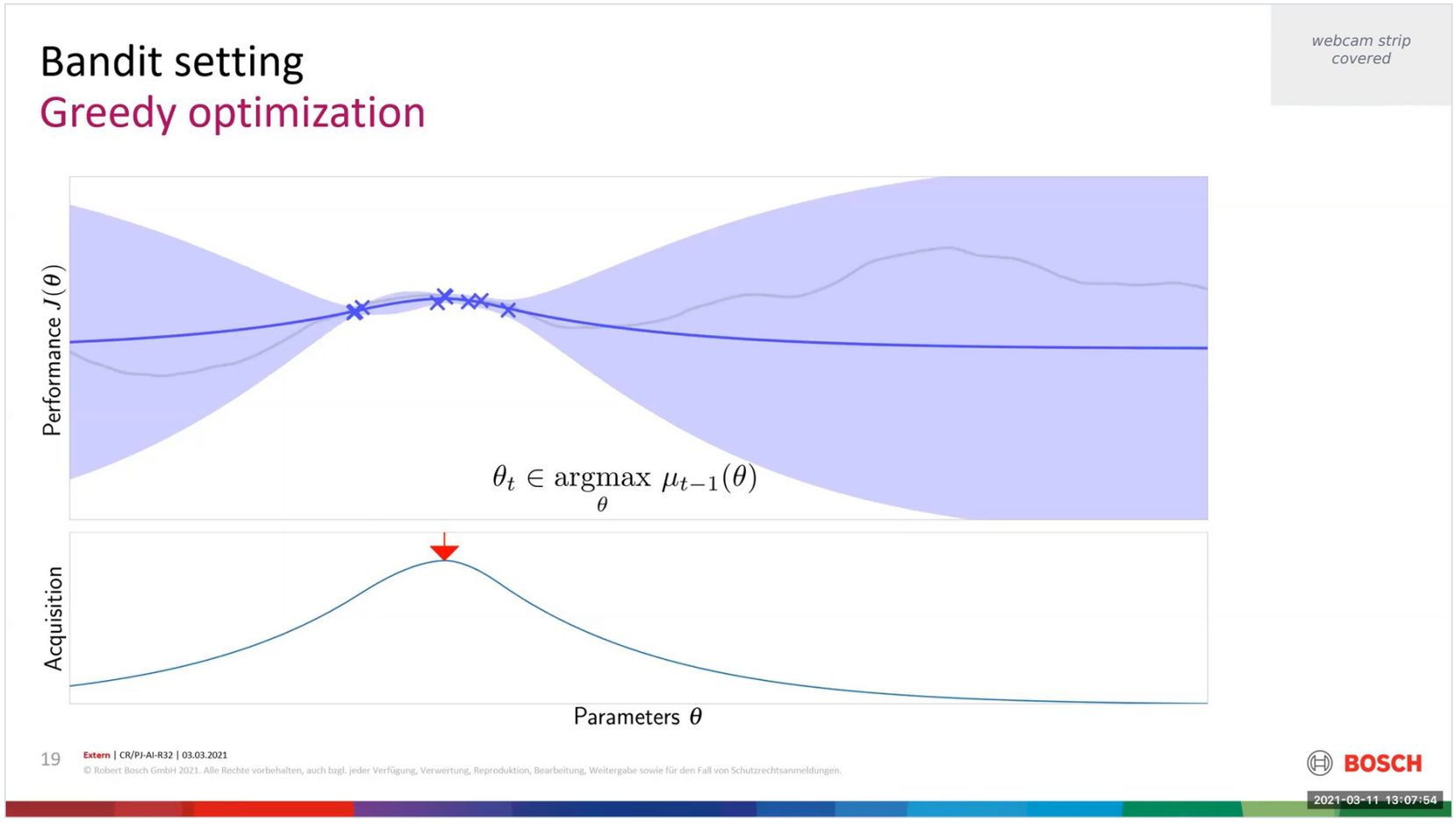
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BANDITS: EXPLORING WITH UNCERTAINTY

First idea: be greedy

▶ 33:46

- **Greedy**: $\theta_t = \text{argmax}_\theta\ \mu_{t-1}(\theta)$. Just maximize the model's current best guess.
- It sounds reasonable: how bad can trusting your best guess be? The next slide tests it.



BANDITS: EXPLORING WITH UNCERTAINTY

Greedy gets stuck

▶ 34:17 · revisited at 35:21, 35:35, 36:18

- Top: the GP model. Bottom: the “acquisition” function being maximized, here just the mean.
- Each new sample lands right next to the previous one. Only tiny randomness in solving the argmax causes any exploration.
- Result: stuck at a **local optimum** on the left. The true maximum is on the right, where the model is very uncertain, but greedy ignores uncertainty and never goes there.

Model-based RL
Exploration schemes

webcam strip covered

Goal: $\max_{\theta \in \mathcal{D}} J(\theta)$

		Exploration
Greedy	$\theta_t = \max_\theta \mu_{t-1}(\theta)$	✘
Optimism	$\theta_t = \max_\pi \mu_{t-1}(\theta) + \beta_t \sigma_{t-1}(\theta)$	✓
Thompson sampling (won't discuss in detail)	$\tilde{J} \sim P(\cdot)$ $\theta_t = \max_\theta \tilde{J}(\theta)$	✓

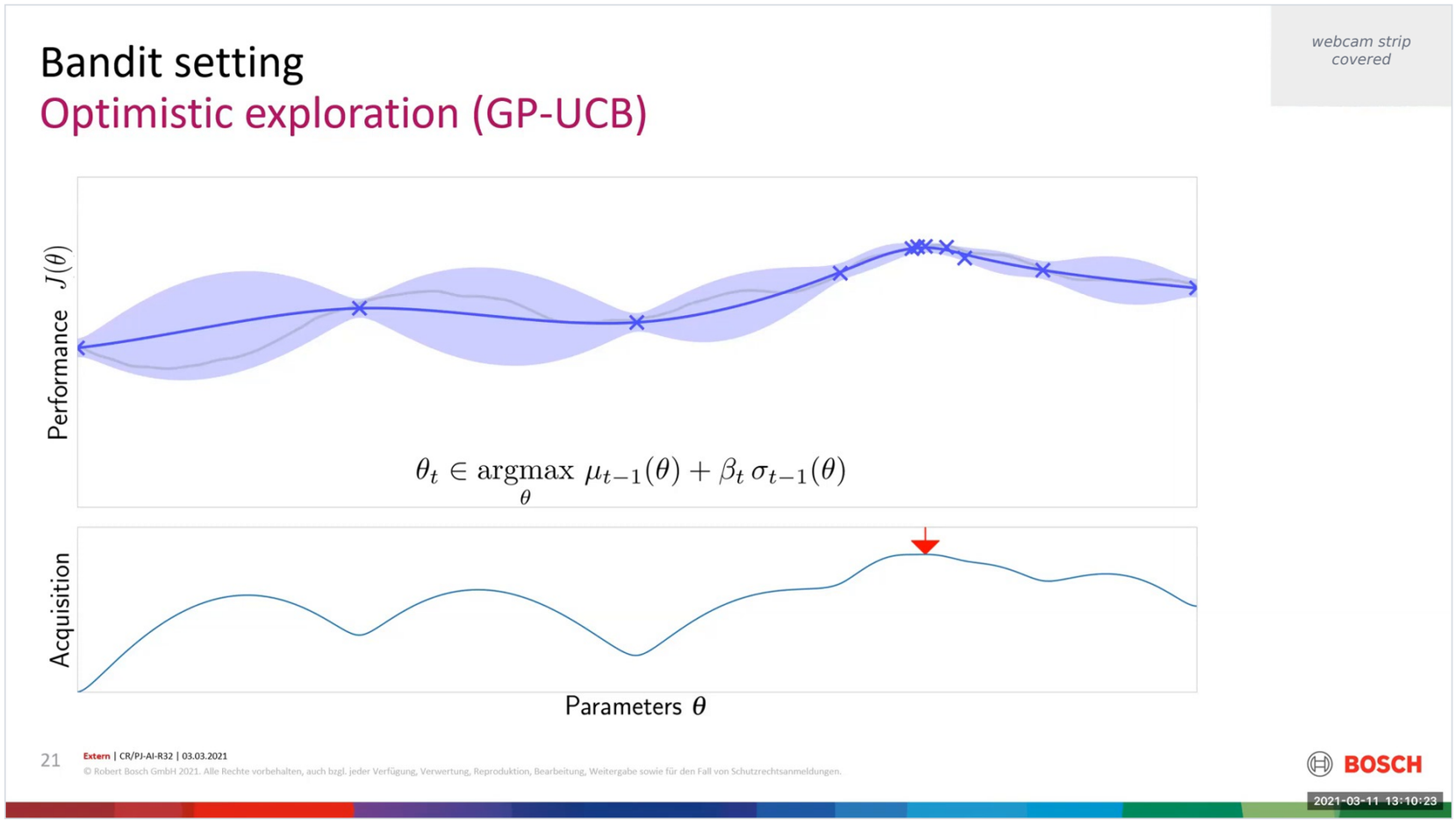
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BANDITS: EXPLORING WITH UNCERTAINTY

Exploration schemes

▶ 35:39

- **Greedy**, θ_t = argmax $\mu_{t-1}(\theta)$: does not explore. ✗
- **Optimism**, θ_t = argmax $\mu_{t-1}(\theta) + \beta_t \sigma_{t-1}(\theta)$: act as if uncertainty works in your favor (pick the top of the band). Provably converges. ✓
- **Thompson sampling**: draw one plausible function $\tilde{J}$ from the model and maximize it. Also converges ✓, but not covered further.



BANDITS: EXPLORING WITH UNCERTAINTY

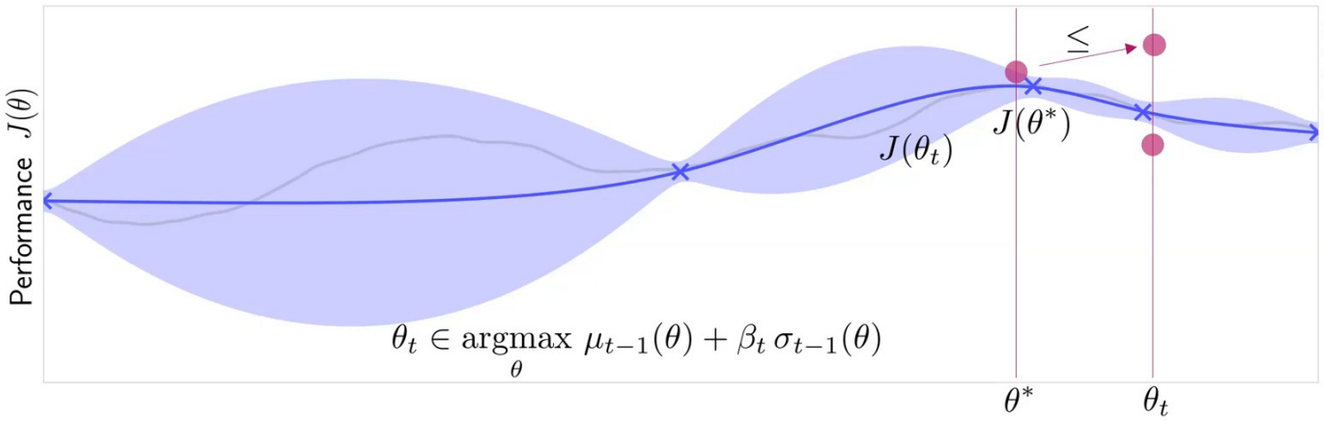
Optimistic exploration (GP-UCB)

▶ 37:05

- GP-UCB picks the top of the confidence band (UCB = upper confidence bound).
- Where data already exists the band is narrow, so it moves on. It seeks out **uncertain** regions.
- Once nothing on the left could beat what has been seen, it concentrates samples near the true optimum on the right. This strategy provably converges.

Bandit setting
Why does optimistic exploration work?

webcam strip covered



Want to bound optimality gap (regret): $J(\theta_t) \geq \mu(\theta_t) - \beta\sigma(\theta_t)$ $J(\theta^*) \leq \mu(\theta^*) + \beta\sigma(\theta^*)$

$J(\theta^*) - J(\theta_t) \leq 2\beta\sigma_{t-1}(\theta_t)$ + extra math to show that variance shrinks $\leq \mu(\theta_t) + \beta\sigma(\theta_t)$

Information-Theoretic Regret Bounds for Gaussian Process Optimization in the Bandit Setting
N. Srinivas, A. Krause, S. Kakade, M. Seeger, IEEE Transactions on Information Theory 2012

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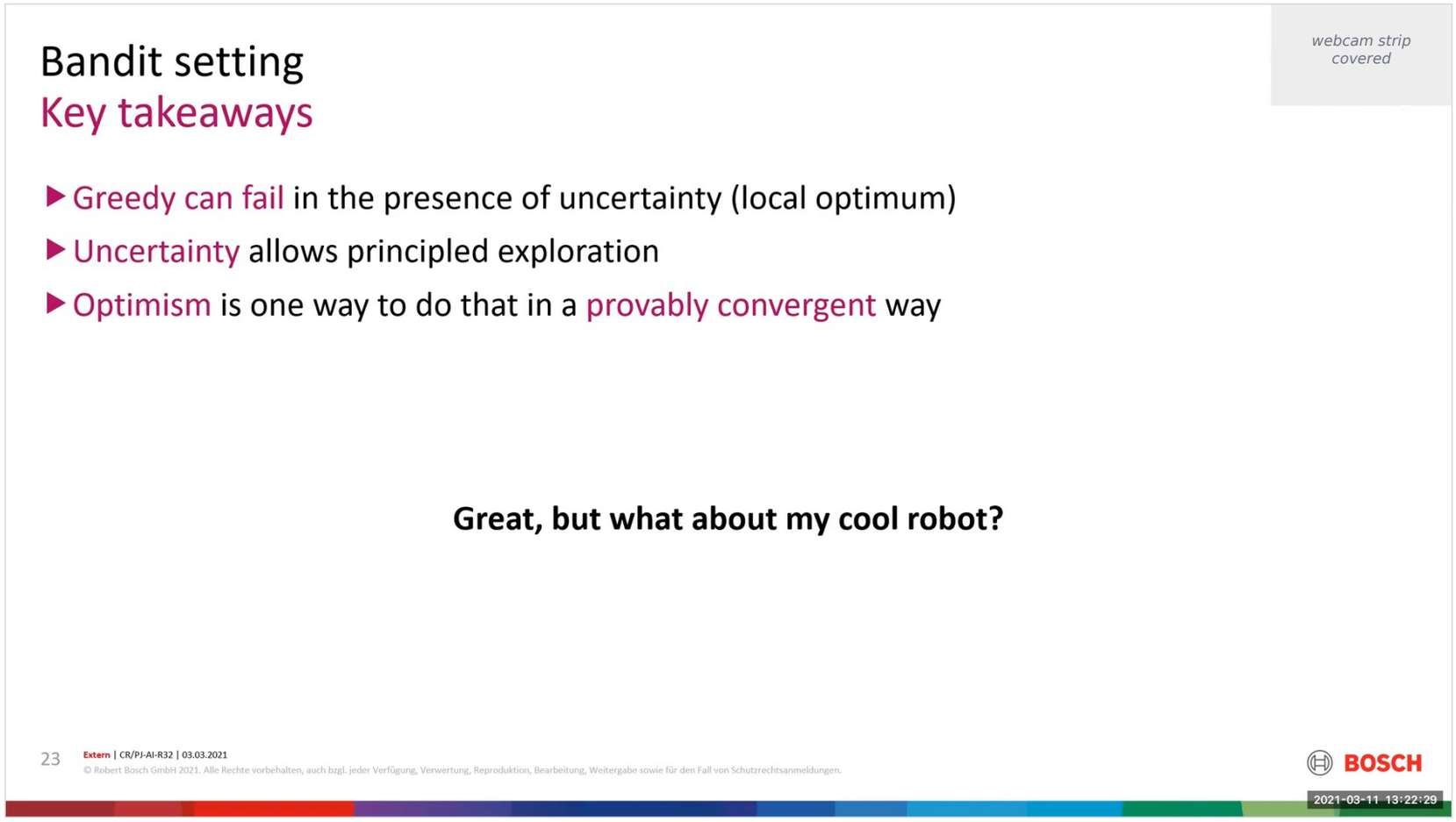
BANDITS: EXPLORING WITH UNCERTAINTY

Why optimism works

▶ 38:08 · revisited at 43:16

- With high probability: $J(\theta_t) \geq \mu(\theta_t) - \beta\sigma(\theta_t)$ (lower bound), and $J(\theta^*) \leq \mu(\theta^*) + \beta\sigma(\theta^*)$ (upper bound).
- Because θ_t maximizes the upper bound: $\mu(\theta^*) + \beta\sigma(\theta^*) \leq \mu(\theta_t) + \beta\sigma(\theta_t)$.
- Subtract to get the per-step **regret**: $J(\theta^*) - J(\theta_t) \leq 2\beta_t\sigma_{t-1}(\theta_t)$. Example: $\beta = 2$, $\sigma = 0.05$ → at most 0.2 below the optimum.
- Every sample shrinks σ, so the gap goes to 0. (The full proof needs extra math on how fast σ shrinks.)

Q&A: *Why does the second inequality hold? Because θ_t was chosen as the max of the upper bound. How to solve that argmax? It runs on the model, not the robot: grid search in low dimensions, or random starts plus gradient ascent; cost grows fast with dimension. Why does β_t grow? The bands must hold at every step of an unknown number of steps, so the failure chance δ is split over rounds (see Srinivas et al.).*



BANDITS: EXPLORING WITH UNCERTAINTY

Bandit takeaways

▶ 49:41

- **Greedy can fail** when there is uncertainty: it gets stuck in a local optimum.
- **Uncertainty** enables principled exploration.
- **Optimism** is one way to do that, with a convergence proof.
- “Great, but what about my cool robot?” Next: back to model-based RL.

Model-based Reinforcement Learning
Problem setup

webcam strip covered

Stochastic system

$$s_{n+1} = f(s_n, a_n) + \omega_n \qquad \omega_n \sim P(\cdot)$$

Noise (aleatoric uncertainty) $\mathbb{E}[\omega_n] = 0$

$$\pi^* \in \max_\pi J(f, \pi)$$

Objective (performance)

$$J(f, \pi) = \mathbb{E}_\omega \left[\sum_{n=0}^{N} r(s_n, \pi(s_n)) \,\middle|\, s_0 \right]$$

Unknown dynamics (epistemic uncertainty)

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BOSCH
2021-03-11 13:22:37

BACK TO MODEL-BASED RL

Model-based RL: same problem, now with a model

▶ 50:14

- Same setup as before: $s_{n+1} = f(s_n, a_n) + \omega_n$, maximize $J(f, \pi)$, dynamics unknown.
- Difference from bandits: we model uncertainty over the **dynamics** f with models $\hat{f}$, and optimize over policies that map state to action.

Model-based Reinforcement Learning
Exploration schemes

webcam strip covered

		Practical?	Exploration
Greedy	$\pi_t = \max_\pi \mathbb{E}_{\tilde{f}\sim P(\cdot)}[J(\tilde{f}, \pi)]$	✓	✘*
Optimism	$\pi_t = \max_{\tilde{f}\in\mathcal{M}_t} \max_\pi J(\tilde{f}, \pi)$	✓	✓
Thompson sampling	$\tilde{f} \sim P(\cdot)$ $\pi_t = \max_\pi J(\tilde{f}, \pi)$	✓	✓

*In general, but not, e.g., for linear-quadratic optimization:
Certainty Equivalence is Efficient for Linear Quadratic Control
H. Mania, S. Tu, B. Recht, NeurIPS, 2019

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BACK TO MODEL-BASED RL

Exploration schemes for model-based RL

▶ 50:22

- **Greedy**: max over π of the expected J over plausible models. Easy, and it is what most RL algorithms do. But bandits are a special case where greedy fails, so it **can't work in general**.
- *Exception: linear dynamics + quadratic cost, where greedy ("certainty equivalence") is efficient (Mania, Tu, Recht, NeurIPS 2019). This may be why control-style greedy methods often work.
- **Optimism**: max over models $\tilde{f} \in M_t$ and policies π of $J(\tilde{f}, \pi)$. Explores, but maximizing over functions is hard.
- **Thompson sampling**: sample a model, optimize for it. Explores, but sampling a non-parametric model is hard.

Model-based Reinforcement Learning

Practical and theoretically sound optimism (H-UCRL)

webcam strip covered

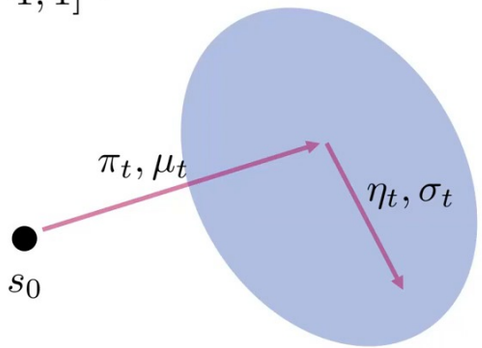
Optimizing in practice: $\pi_t = \max_{\tilde{f} \in \mathcal{M}} \max_{\pi} J(\tilde{f}, \pi)$

Well-calibrated model: $|f(\cdot) - \mu_t(\cdot)| \leq \beta_t \sigma_t(\cdot)$

Re-parameterize epistemic: $\tilde{f} = \mu_t(\cdot) + \beta_t \sigma_t(\cdot) \eta(\cdot) \qquad \eta(\cdot) \in [-1, 1]^{d_s}$

Optimistic policy: $\pi_t, \eta_t = \max_{\pi, \eta} J(\tilde{f}, (\pi, \eta))$

"optimistic dynamics"



Efficient Model-Based Reinforcement Learning through Optimistic Policy Search and Planning
S. Curi*, F. Berkenkamp*, A. Krause, NeurIPS 2020

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BACK TO MODEL-BASED RL

H-UCRL: optimism made practical

▶ 53:15

- Use calibration: the true f is within $\mu_t \pm \beta_t \sigma_t$. So write any plausible model as $\tilde{f} = \mu_t + \beta_t \sigma_t \cdot \eta$, with $\eta \in [-1, 1]^d$ per state dimension.
- Treat η as an extra, **"hallucinated" control input** that can push each predicted step anywhere inside the uncertainty band.
- Optimize π and η together: $(\pi_t, \eta_t) = \text{argmax } J(\tilde{f}, (\pi, \eta))$. This yields "optimistic dynamics" and an optimistic policy.
- A hard max over models becomes an ordinary policy optimization with a larger action space. Curi, Berkenkamp, Krause, NeurIPS 2020.

BACK TO MODEL-BASED RL

H-UCRL on an inverted pendulum

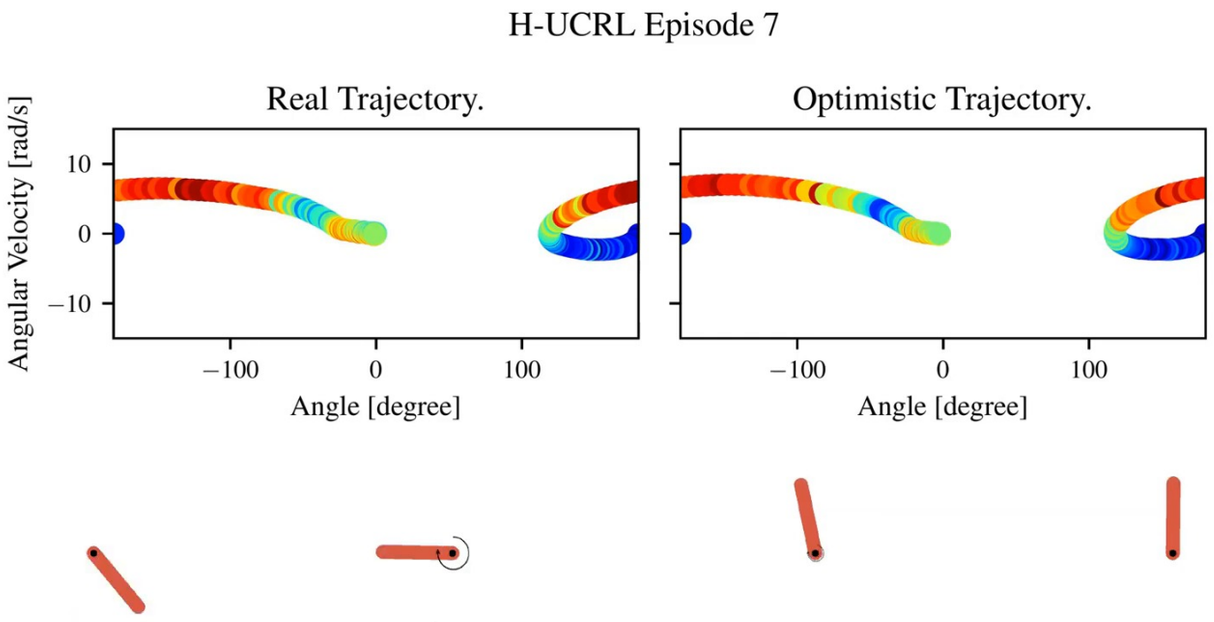
▶ 55:22 · revisited at 56:27, 56:44, 56:56

- Task: swing a pendulum up from hanging down (−180°). Left: the real trajectory. Right: the optimistic plan.
- Episode 1: the plan goes straight up with no swing, “ignoring physics”, and the real pendulum barely moves. But the data it collects is informative.
- Episode 3: the plan now includes a swing-up. By episodes 5–7 plan and reality match and the pendulum balances.
- The uncertainty here comes from an **ensemble of neural networks**, not a GP.

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Model-based Reinforcement Learning

H-UCRL: Inverted pendulum



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Half-cheetah: when greedy breaks

▶ 57:10

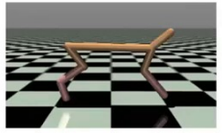
- Return vs. episode on the half-cheetah benchmark, with action penalty 0, 0.1 and 1.0.
- No penalty: greedy also learns, just more slowly; H-UCRL is a bit better. (PSRL, a Thompson-style method, is only approximate here, so ignore it.)
- With action costs, greedy stalls and at 1.0 barely learns. H-UCRL still learns.
- Why: greedy treats epistemic uncertainty as noise, so the system looks hopelessly random. With action costs, “do nothing” then looks best.

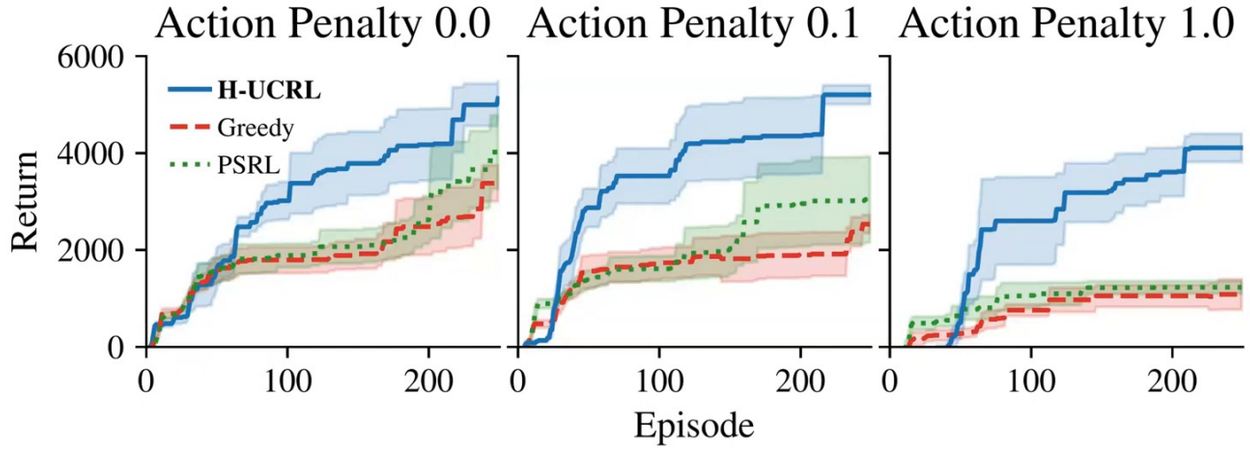
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Model-based Reinforcement Learning

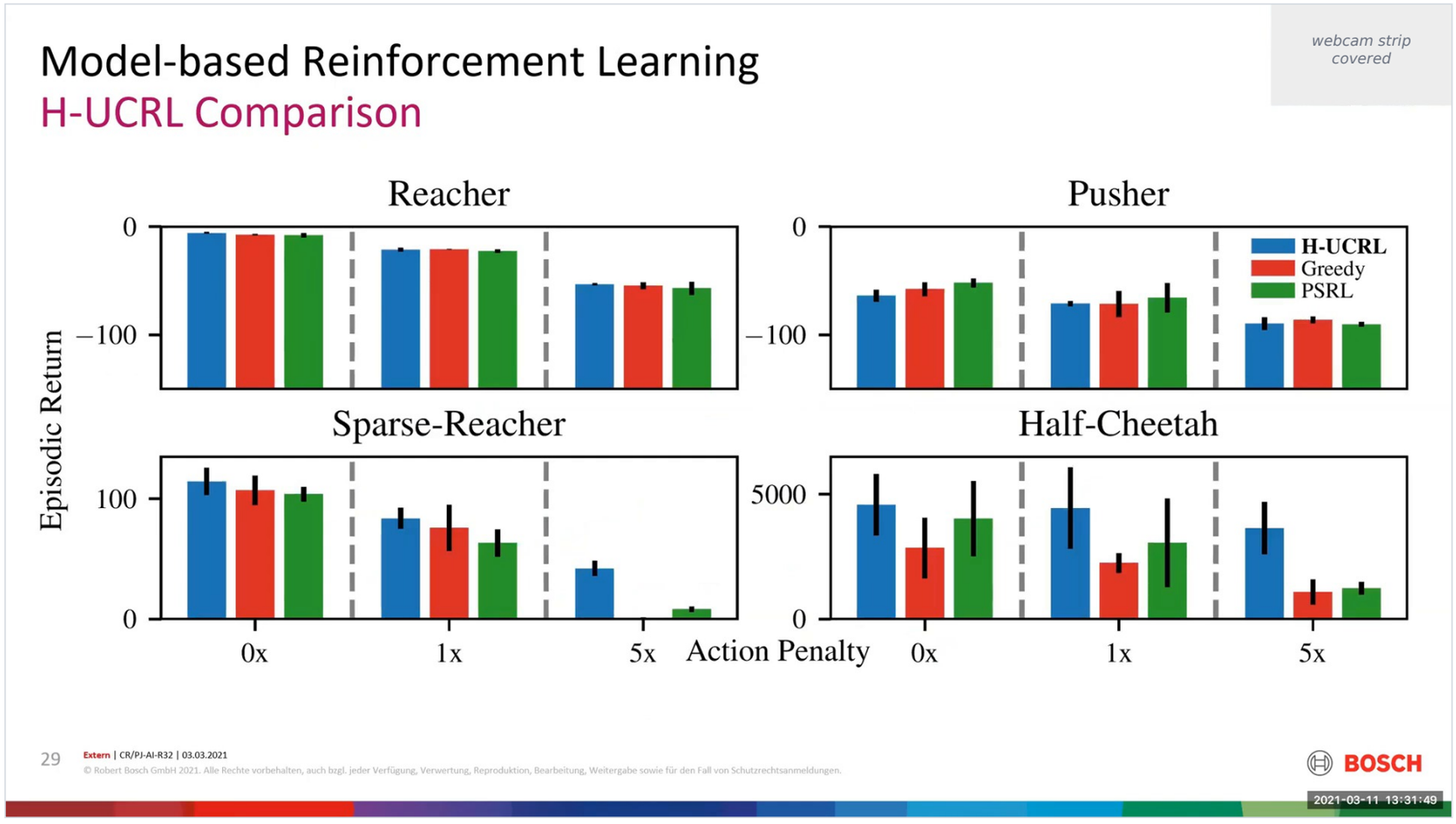
H-UCRL Comparison

Half-cheetah





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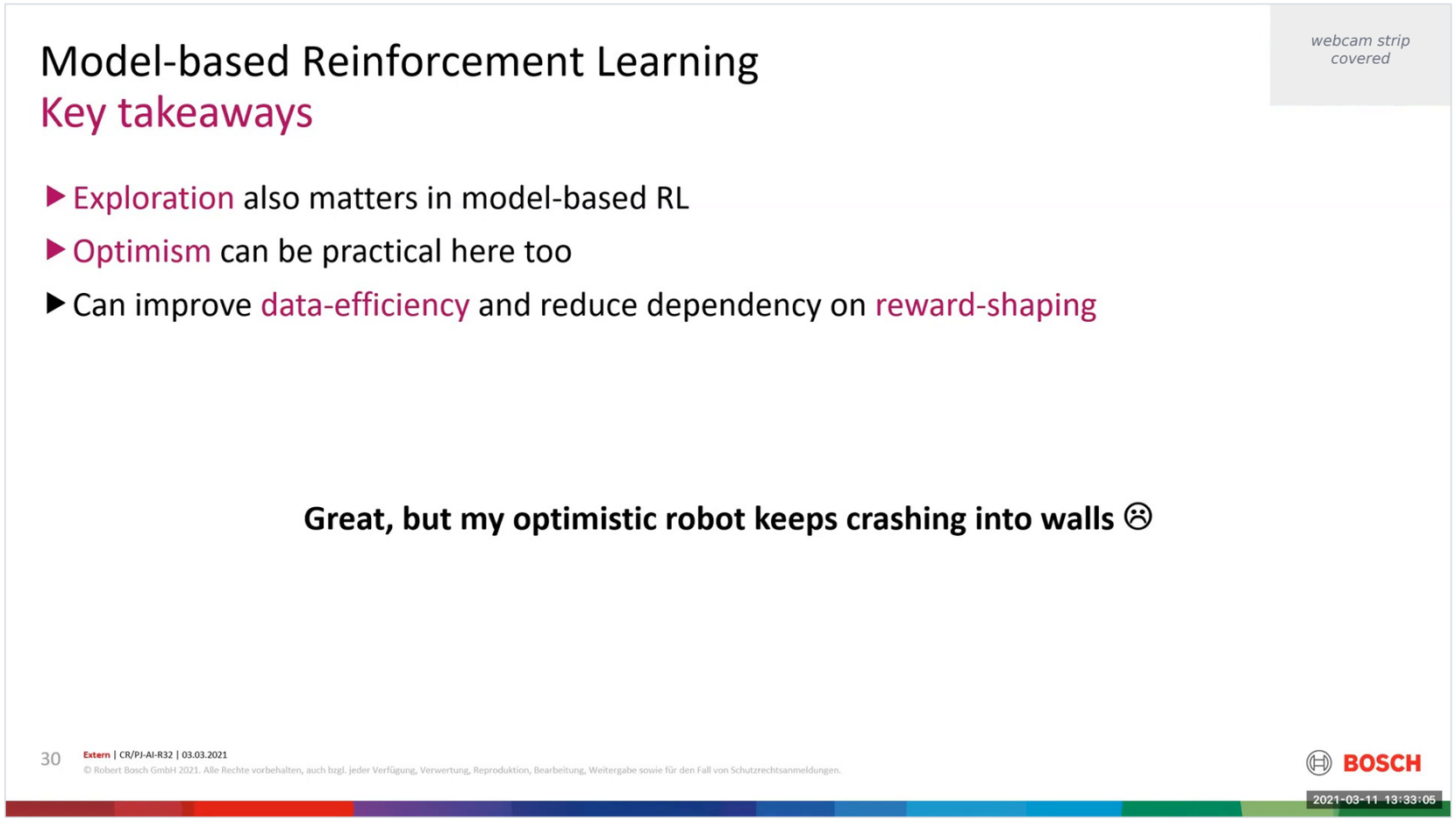


BACK TO MODEL-BASED RL

Same trend on more tasks

▶ 59:29

- Final return on Reacher, Pusher, Sparse-Reacher and Half-Cheetah at action penalties 0×, 1× and 5×.
- As the penalty grows, greedy (red) falls behind, most clearly on Sparse-Reacher and Half-Cheetah, while H-UCRL (blue) holds up.

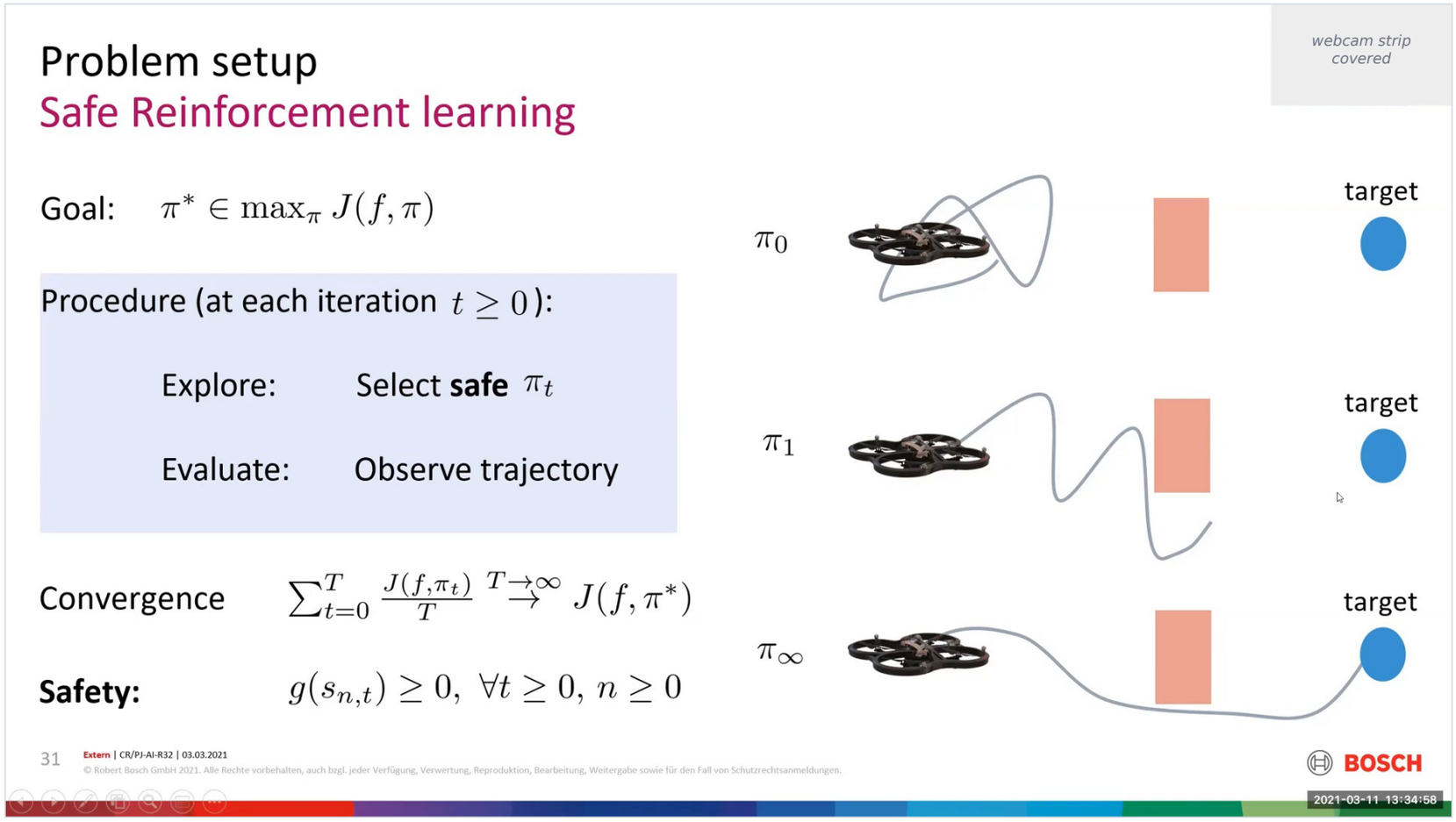


BACK TO MODEL-BASED RL

Model-based RL takeaways

▶ 59:36 · revisited at 1:01:02

- **Exploration** matters in model-based RL too.
- **Optimism** can be practical here (H-UCRL).
- It improves **data-efficiency** and reduces the need for **reward shaping** (hand-tuned rewards often used to hide the action-cost problem).
- “But my optimistic robot keeps crashing into walls.” Optimism assumes unknown things are good, e.g. that walls aren’t there. Safety hasn’t been handled yet.



ADDING SAFETY

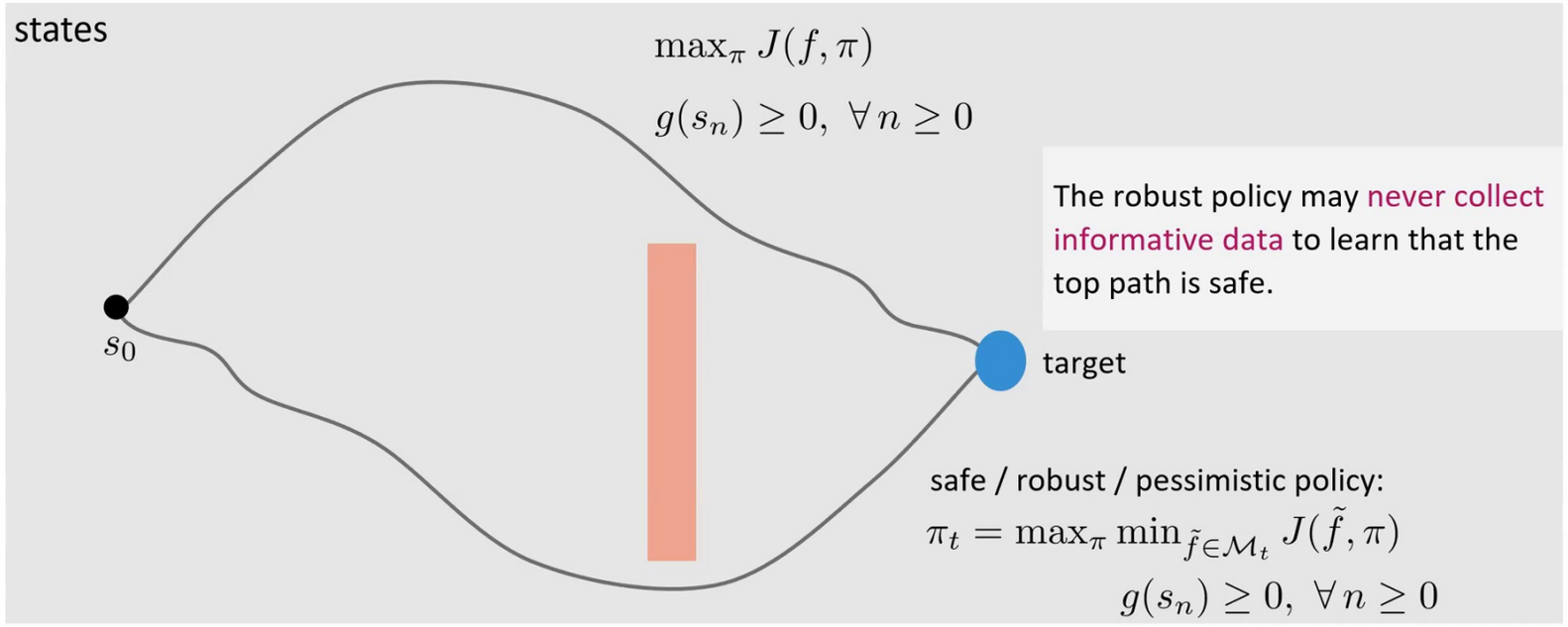
Safe reinforcement learning

▶ 1:00:50

- Same loop, but every policy tried must be **safe**, and we still want convergence to the best safe policy.
- Safety constraint: $g(s_{n,t}) \geq 0$ for every step n of every iteration t, i.e. during learning, not only at the end.
- Example: the quadrotor must reach the target without ever hitting the wall (red block) on any try.
- (Deck page 48 repeats this slide; it flashed by for under a second at 78:08.)

Problem setup
Why is this challenging?

webcam strip covered



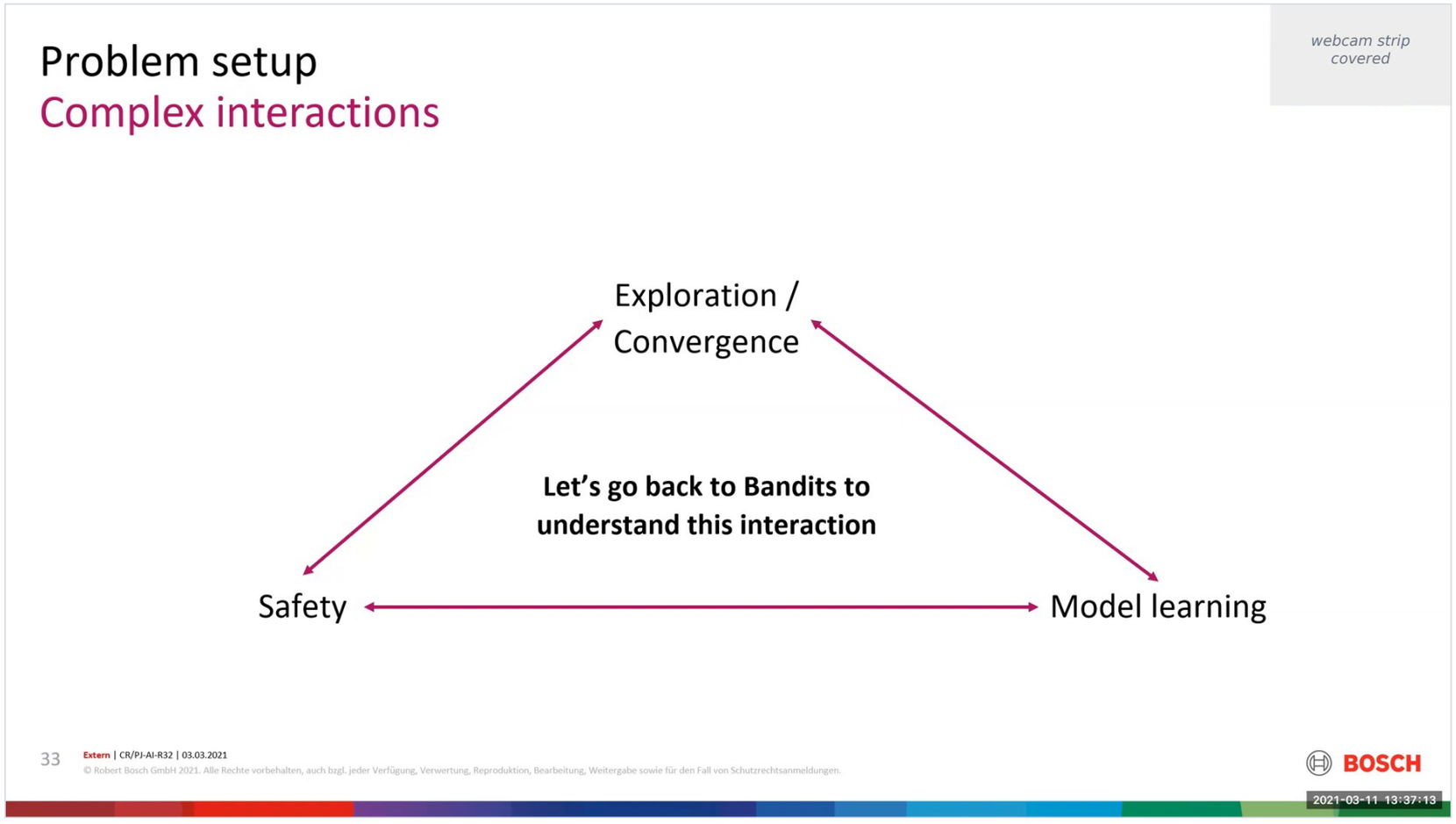
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ADDING SAFETY

Why safe exploration is hard

▶ 1:02:43

- A **robust (pessimistic)** policy plans for the worst plausible model: $\pi_t = \text{argmax}_\pi \min_{\tilde{f} \in M_t} J(\tilde{f}, \pi)$, keeping $g \geq 0$ for all models. That is what people usually call a “safe” policy.
- It may always take the long, safe route (bottom), so it **never collects data** showing the shorter top route is also safe.
- So it may never find the best safe policy. Simply combining safety with a standard exploration method usually fails.

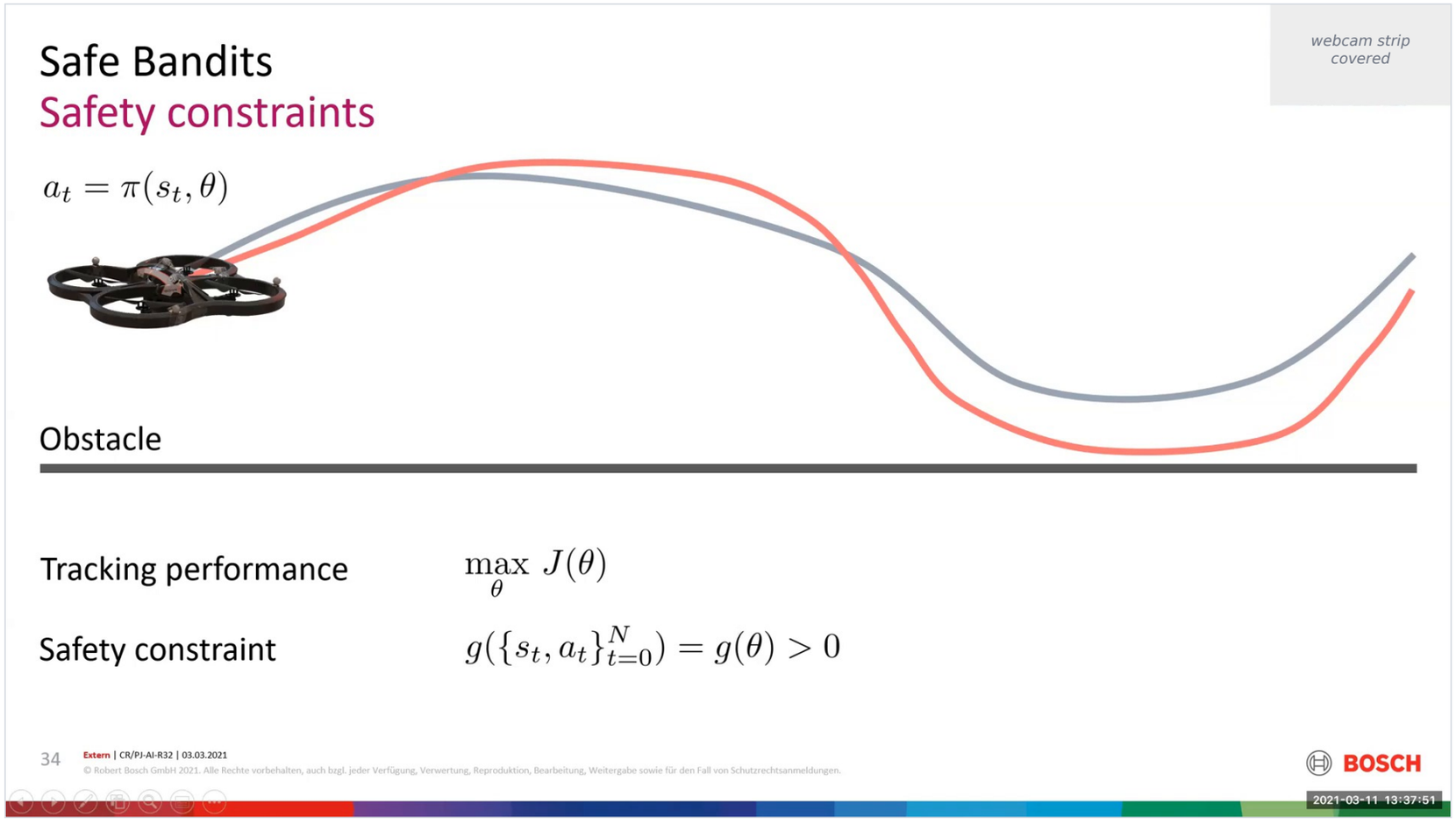


ADDING SAFETY

The triangle, now with safety

▶ 1:04:20

- Safety needs pessimism, which limits **where** we can explore.
- That limits what the model can learn, and the model in turn decides what counts as safe.
- To understand this, go back to bandits, where the same problems appear.



SAFE BANDITS

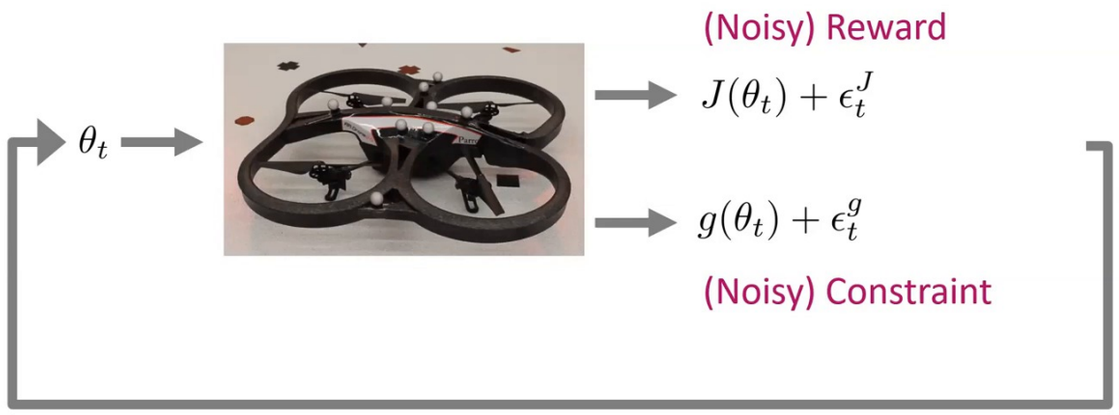
Safe bandits: a safety constraint

▶ 1:04:58

- Same one-step setup: fly with parameters θ and measure tracking performance J(θ).
- Add a safety function g(θ) that must stay above 0. Example: g = the closest distance to the obstacle along the flight. Positive means no crash.

Safe Bandits
Setup

webcam strip covered



Goal: $\max_\theta J(\theta)$ s.t. $g(\theta) \geq 0$

Safety: $g(\theta_t) \geq 0$ for all t with probability $\geq 1 - \delta$

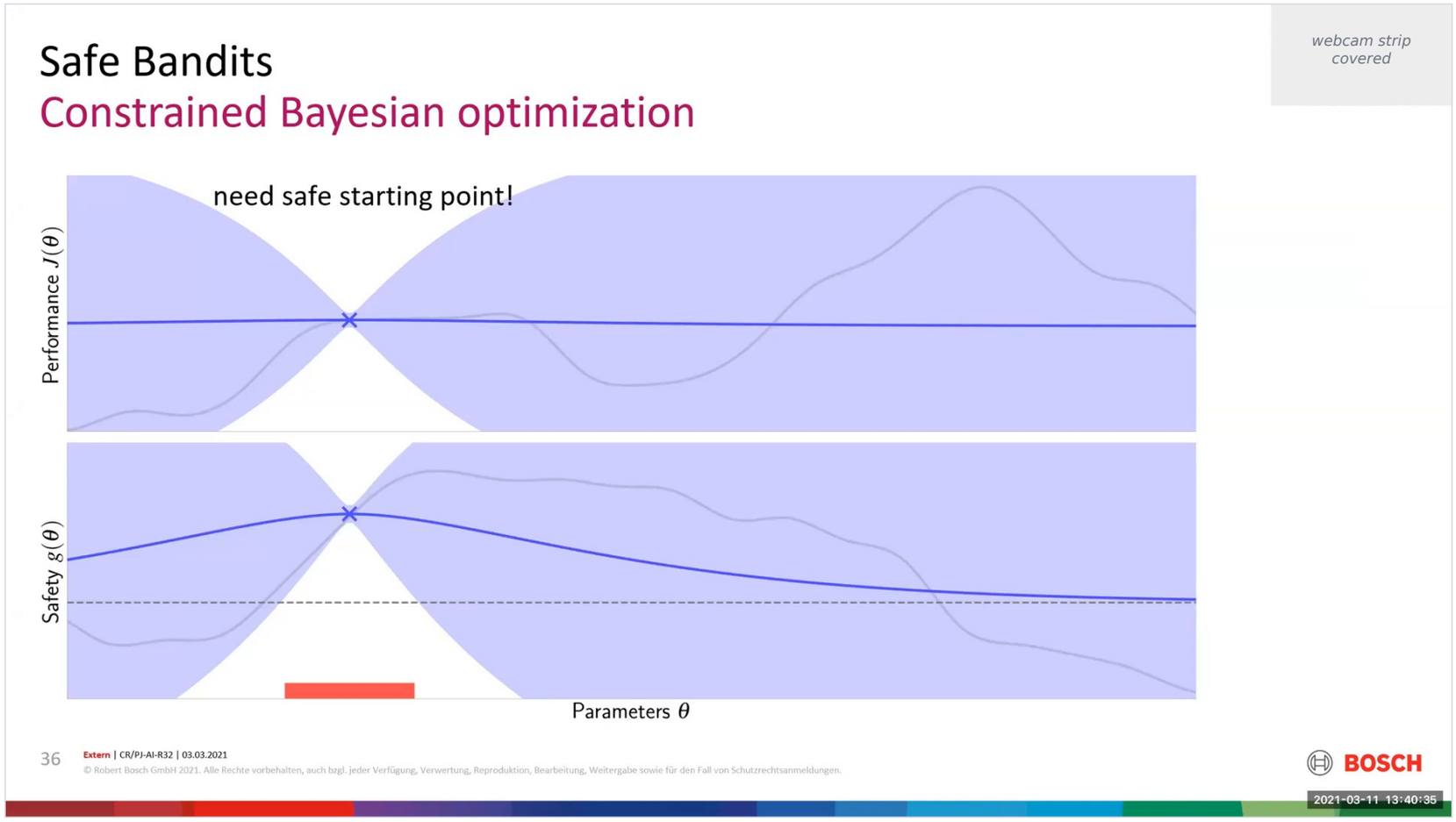
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SAFE BANDITS

Safe bandits: setup

▶ 1:05:36

- Each round: pick θ_t, observe a noisy reward $J(\theta_t) + \varepsilon$ and a noisy constraint value $g(\theta_t) + \varepsilon$.
- Goal: max $J(\theta)$ subject to $g(\theta) \geq 0$.
- Safety: $g(\theta_t) \geq 0$ at **every** round, with probability $\geq 1 - \delta$. No unsafe try is allowed, even while learning.



SAFE BANDITS

Constrained Bayesian optimization

▶ 1:06:05 · revisited at 1:07:22

- Two GP models: one for performance J, one for safety g (dashed line = threshold 0).
- With no data the models know nothing, so you need a **known safe starting point**: prior knowledge, like someone first showing you how to make the drone hover.
- After that one point, every θ whose **lower** confidence bound on g is above 0 is safe with high probability. That is the red bar, the current **safe set**.
- In this one-step setting, checking safety is easy: is the lower bound of g ≥ 0?

Safe Bandits
How to optimize?

webcam strip covered

Idea for exploration:

Safe optimism:

$$\theta_t = \max_{\theta} \mu_{t-1}(\theta) + \beta_t \sigma_{t-1}(\theta) \quad \text{optimism on objective}$$

$$\text{s.t. } P(g(\theta) \geq 0) \geq 1 - \delta \quad \text{pessimism on safety}$$

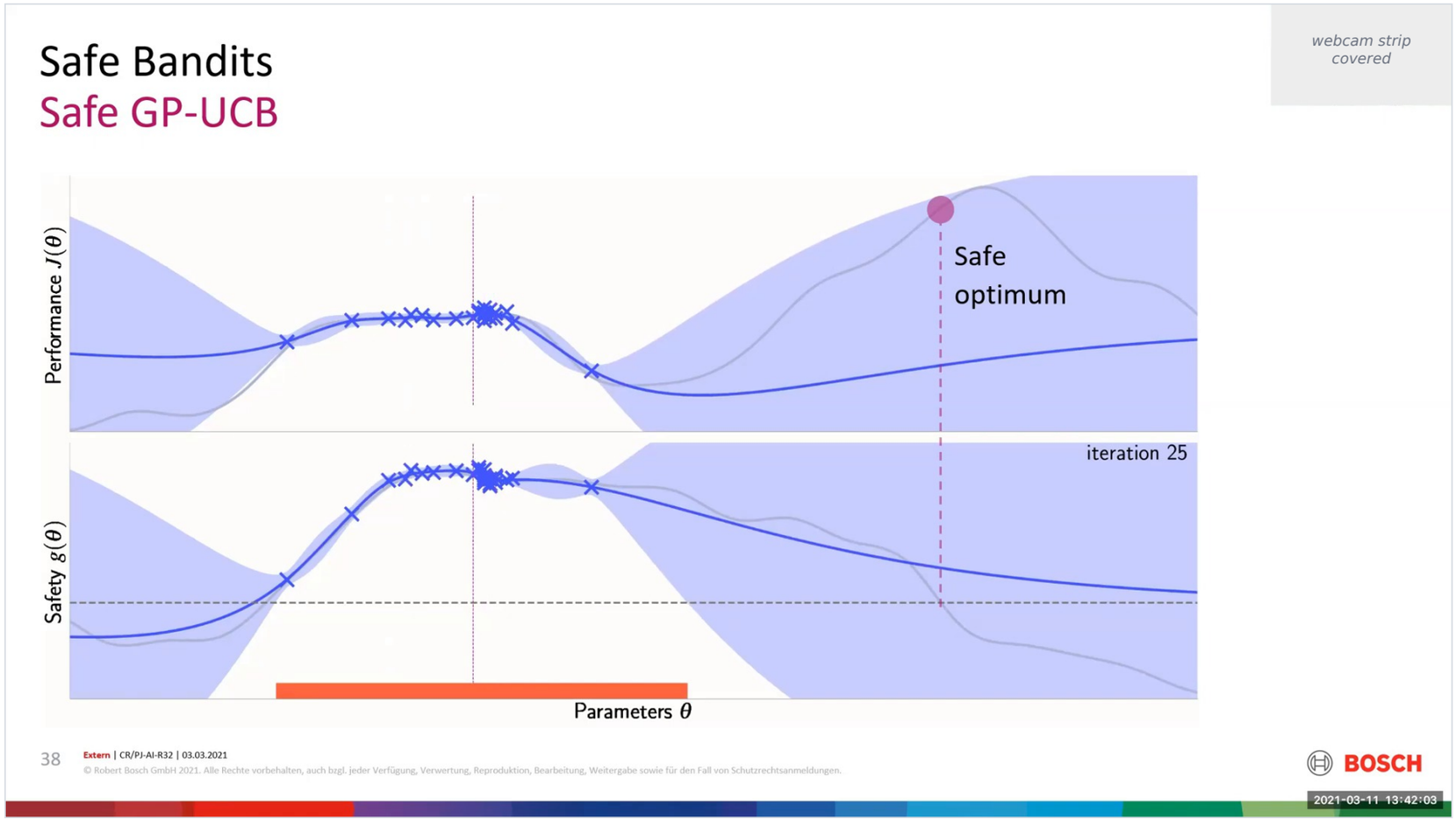
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SAFE BANDITS

How to optimize? Try “safe optimism”

▶ 1:08:20

- Natural idea: be **optimistic on the objective** and **pessimistic on safety**.
- $\theta_t = \text{argmax } \mu_{t-1}(\theta) + \beta_t \sigma_{t-1}(\theta)$, subject to $P(g(\theta) \geq 0) \geq 1 - \delta$.
- This is the common recipe in practice: take an RL method with exploration guarantees and add safety constraints. Does it still work?

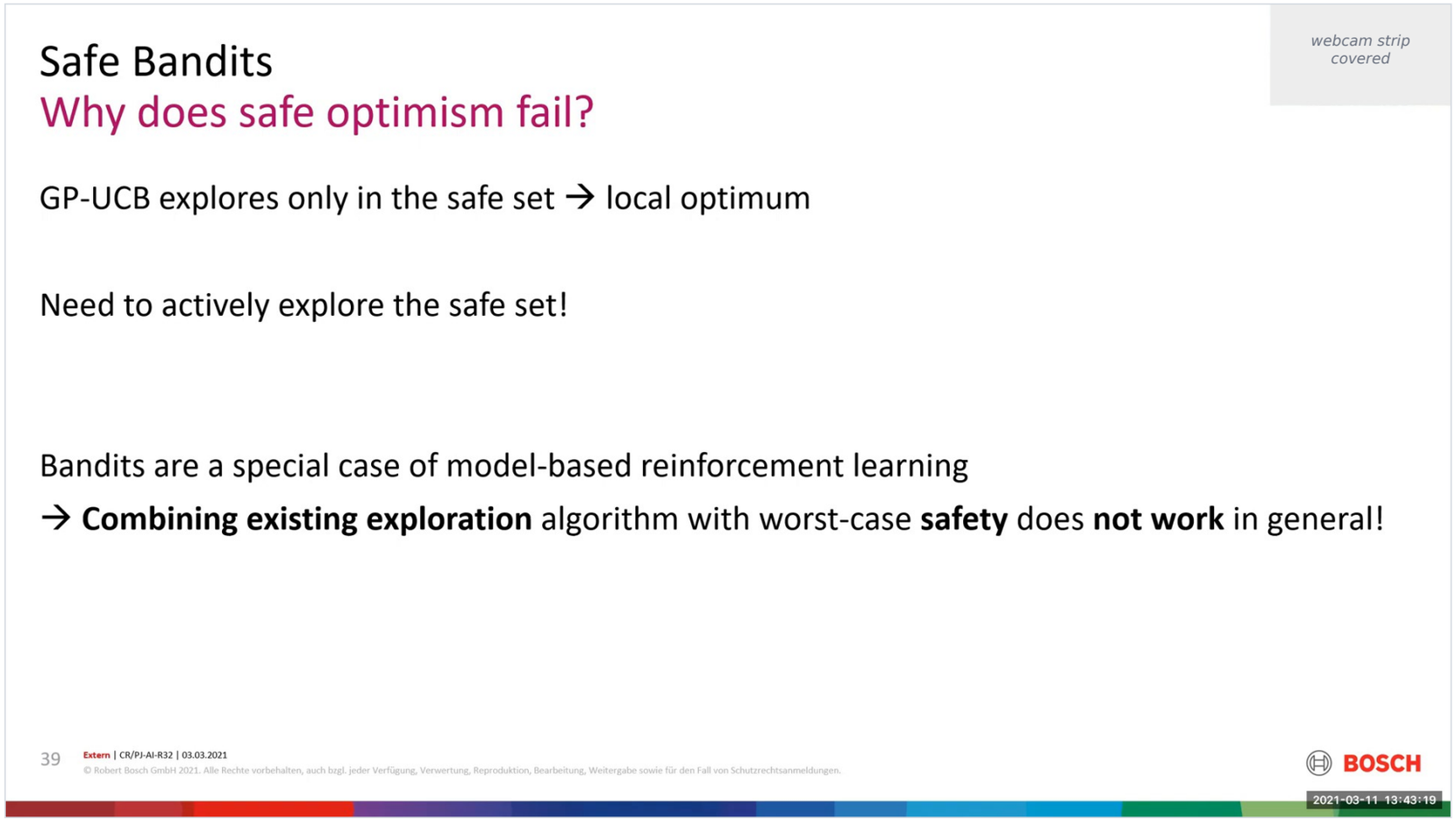


SAFE BANDITS

Safe GP-UCB gets stuck

▶ 1:09:09

- Iterations 1 → 25: it looks promising at first, then piles samples onto a **local optimum** inside the starting safe region.
- The **safe optimum** on the right (pink dot) is never reached. The safe set (red bar) stops growing.
- Reason: it only samples where it is already sure things are safe, so nothing pushes it to learn whether nearby regions are safe too.

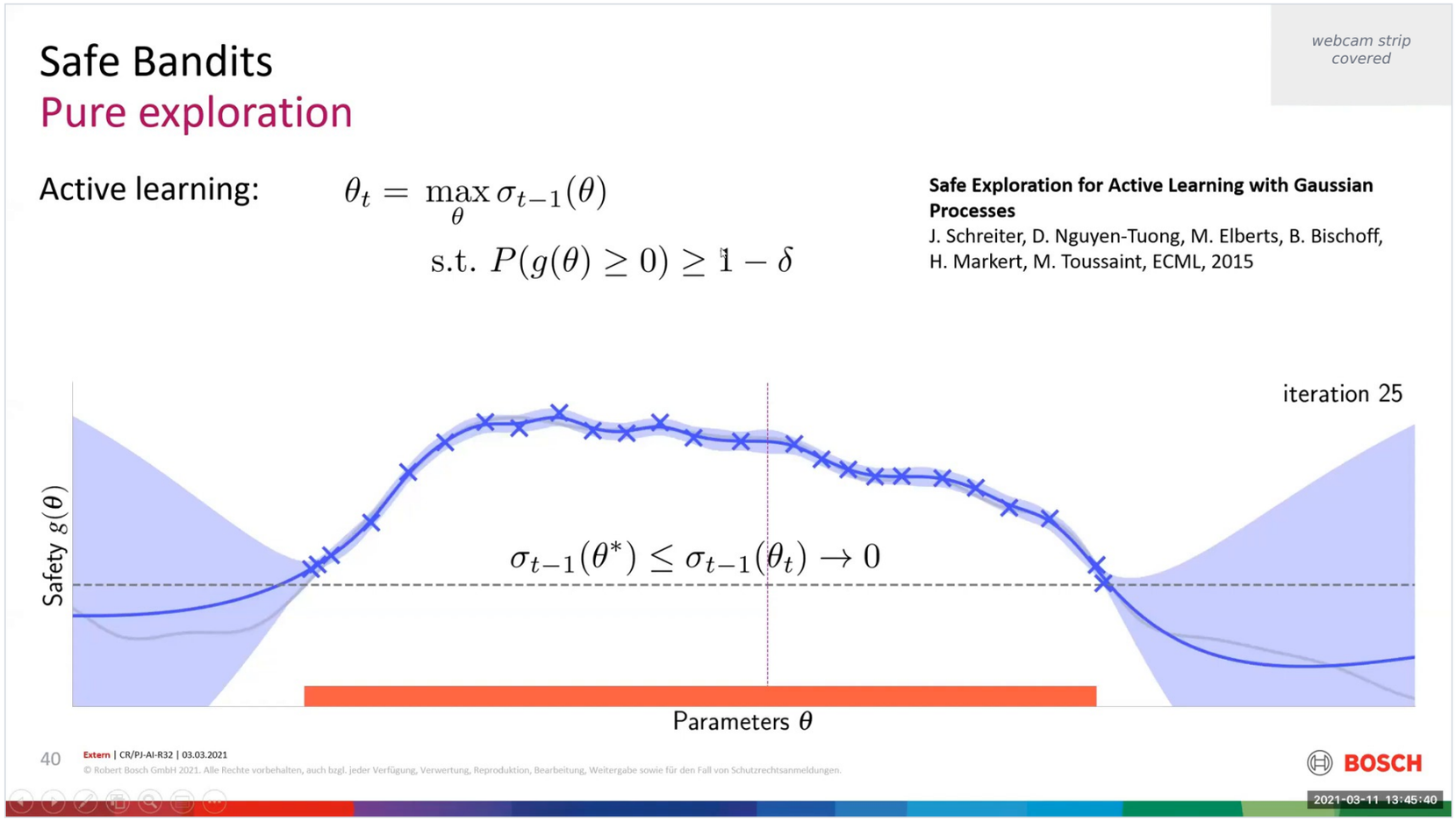


SAFE BANDITS

Why safe optimism fails

▶ 1:10:01

- GP-UCB explores only inside the safe set, so it ends at a local optimum.
- You must **actively explore the safe set**, i.e. deliberately learn about the safety constraint.
- Bandits are a special case of model-based RL, so combining existing exploration algorithms with worst-case safety **does not work in general**. (Linear-quadratic systems are again an exception.)



SAFE BANDITS

Pure exploration (active learning)

▶ 1:11:04 · revisited at 1:12:24, 1:12:58, 1:13:07

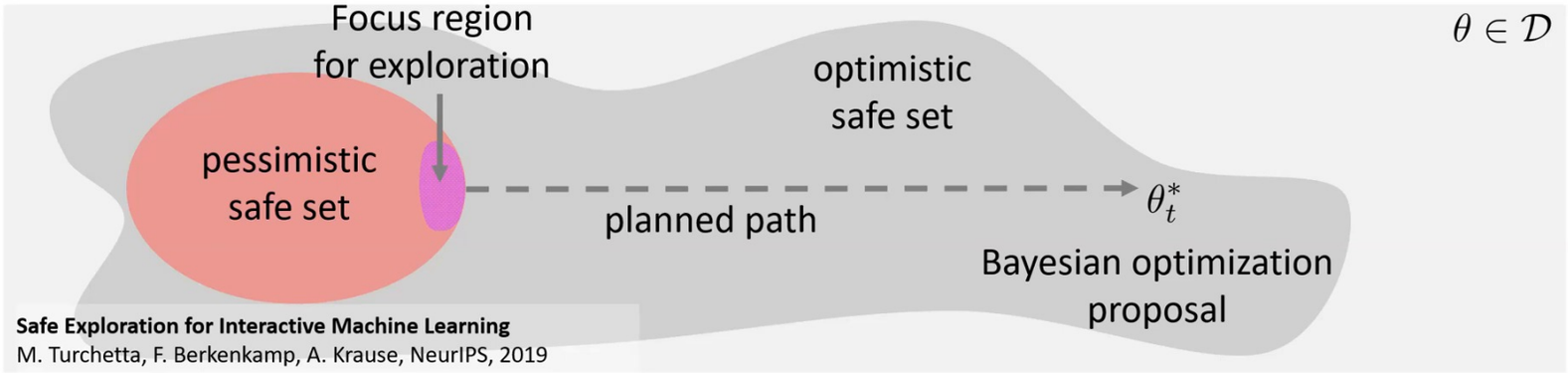
- Ignore the objective: $\theta_t = \text{argmax } \sigma_{t-1}(\theta)$ subject to $P(g(\theta) \geq 0) \geq 1 - \delta$. Sample the **most uncertain safe** point (Schreiter et al., ECML 2015).
- The most uncertain points sit at the edge of the safe set, so the set keeps growing (red bar, iterations 21 → 25).
- It converges because $\sigma(\theta^*) \leq \sigma(\theta_t) \to 0$. But it is data-hungry: it learns about regions you don't care about.
- Speaker: outside special cases, this is the only approach currently known to work. Doing better is open.

Safe Bandits

Goal-oriented safe exploration

Idea: Combine exploration with optimization

Bayesian optimization on an optimistic safe set, then explore to learn about safety.



Safe Exploration for Interactive Machine Learning
M. Turchetta, F. Berkenkamp, A. Krause, NeurIPS, 2019

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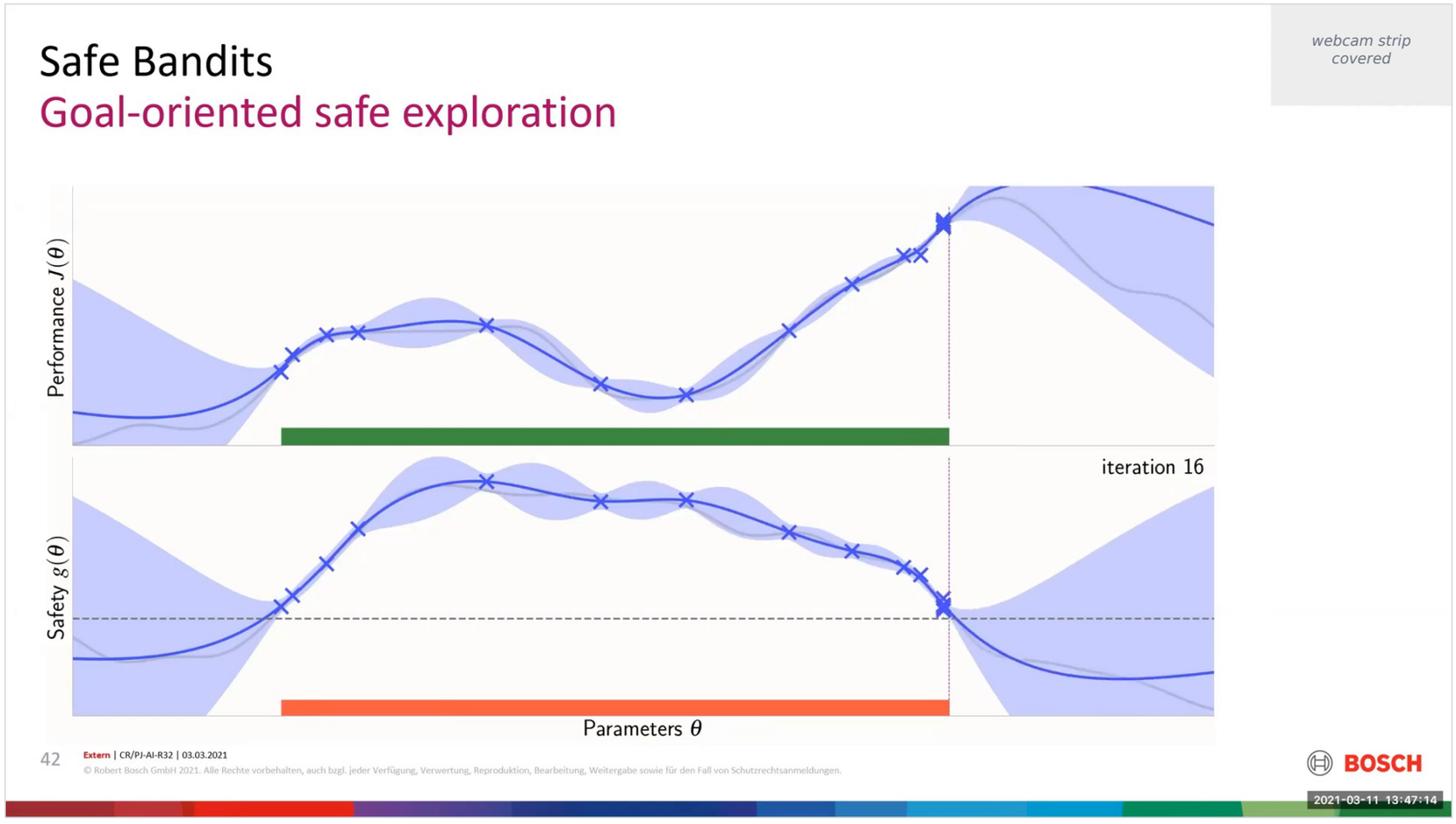
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SAFE BANDITS

Goal-oriented safe exploration

▶ 1:13:25

- Idea: combine exploration with optimization.
- Run Bayesian optimization on the **optimistic** safe set (everything that *might* be safe) to get a target θ^*_t.
- Then explore for safety only near the edge of the **pessimistic** safe set, along a planned path toward that target (Turchetta, Berkenkamp, Krause, NeurIPS 2019).
- Speaker's view: it works, but it is a hacky mix of "explore everywhere" and directed search, not the final answer.



SAFE BANDITS

Goal-oriented safe exploration in action

▶ 1:14:41

- Same two-model plot as before (performance on top, safety below), iterations 1 → 16.
- The safe set (red bar) keeps growing to the right, toward the promising region, instead of stalling as safe GP-UCB did.
- Samples follow and reach the region of the safe optimum: the algorithm actively explores and converges.
- (Deck page 42: an animation that never stood still, so it was easy to miss.)

Safe Bandits
References

webcam strip covered

Safe Exploration for Optimization with Gaussian Processes
Y. Sui, A. Gotovos, J.W. Burdick, A. Krause, ICML, 2015

Stagewise Safe Bayesian Optimization with Gaussian Processes
Y. Sui, V. Zhuang, J.W. Burdick, Y. Yue, ICML, 2018

Safe Exploration for Interactive Machine Learning
M. Turchetta, F. Berkenkamp, A. Krause, NeurIPS, 2019

Overview at: https://www.youtube.com/watch?v=sMfPRLtrob4

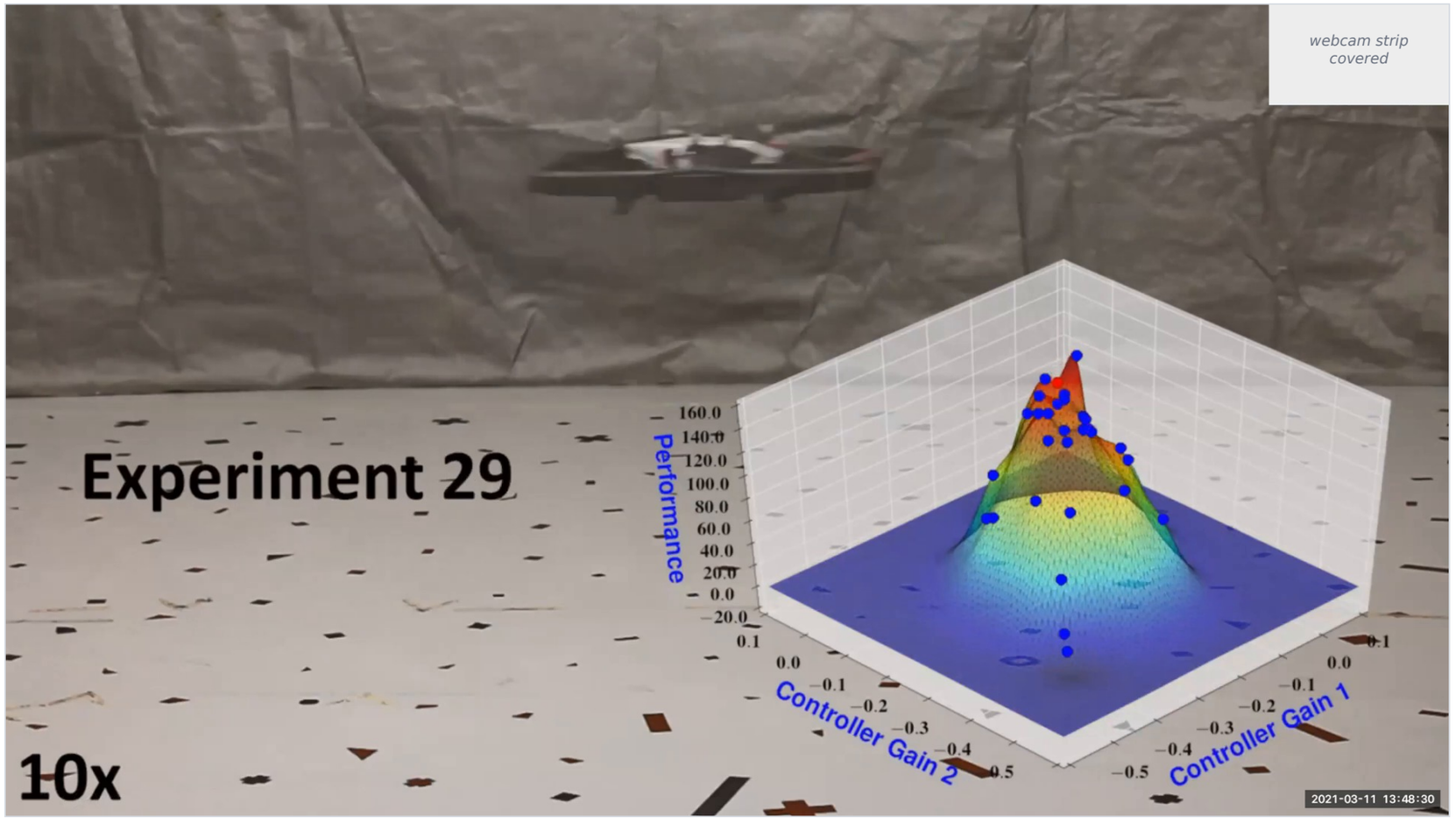
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SAFE BANDITS

References: safe bandits

▶ 1:15:00

- Sui, Gotovos, Burdick, Krause, ICML 2015: safe exploration for optimization with GPs (SafeOpt).
- Sui, Zhuang, Burdick, Yue, ICML 2018: stagewise safe Bayesian optimization.
- Turchetta, Berkenkamp, Krause, NeurIPS 2019: safe exploration for interactive machine learning.
- His summary: we're "not quite there". These methods rely on learning about the function everywhere in the worst case.

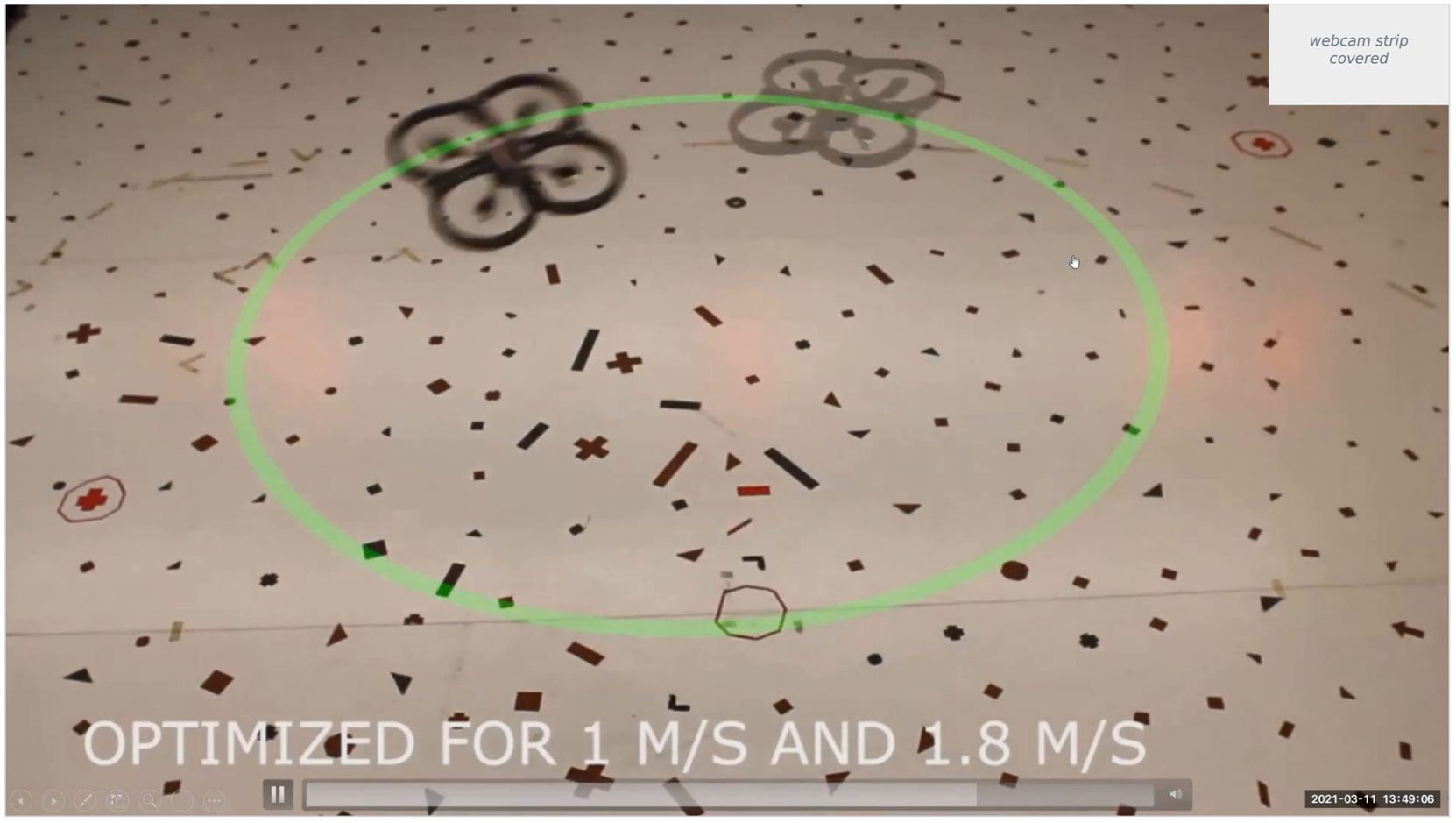


SAFE BANDITS

Video: safe tuning of a real quadrotor

▶ 1:15:26

- From Berkenkamp, Schoellig, Krause (2016), Bayesian optimization with safety constraints. Full-screen video; deck pages 44–46 were never visible and likely hold it.
- Task: fly left to right as fast as possible by tuning two controller gains. Bottom right: the GP model of performance over the gains.
- Safety constraint: performance must not drop below a threshold.
- Over the experiments shown (1 → 29) it picks gains that are both safe and informative. It ends near optimal without a single crash, though crash-causing gains existed in the search space.

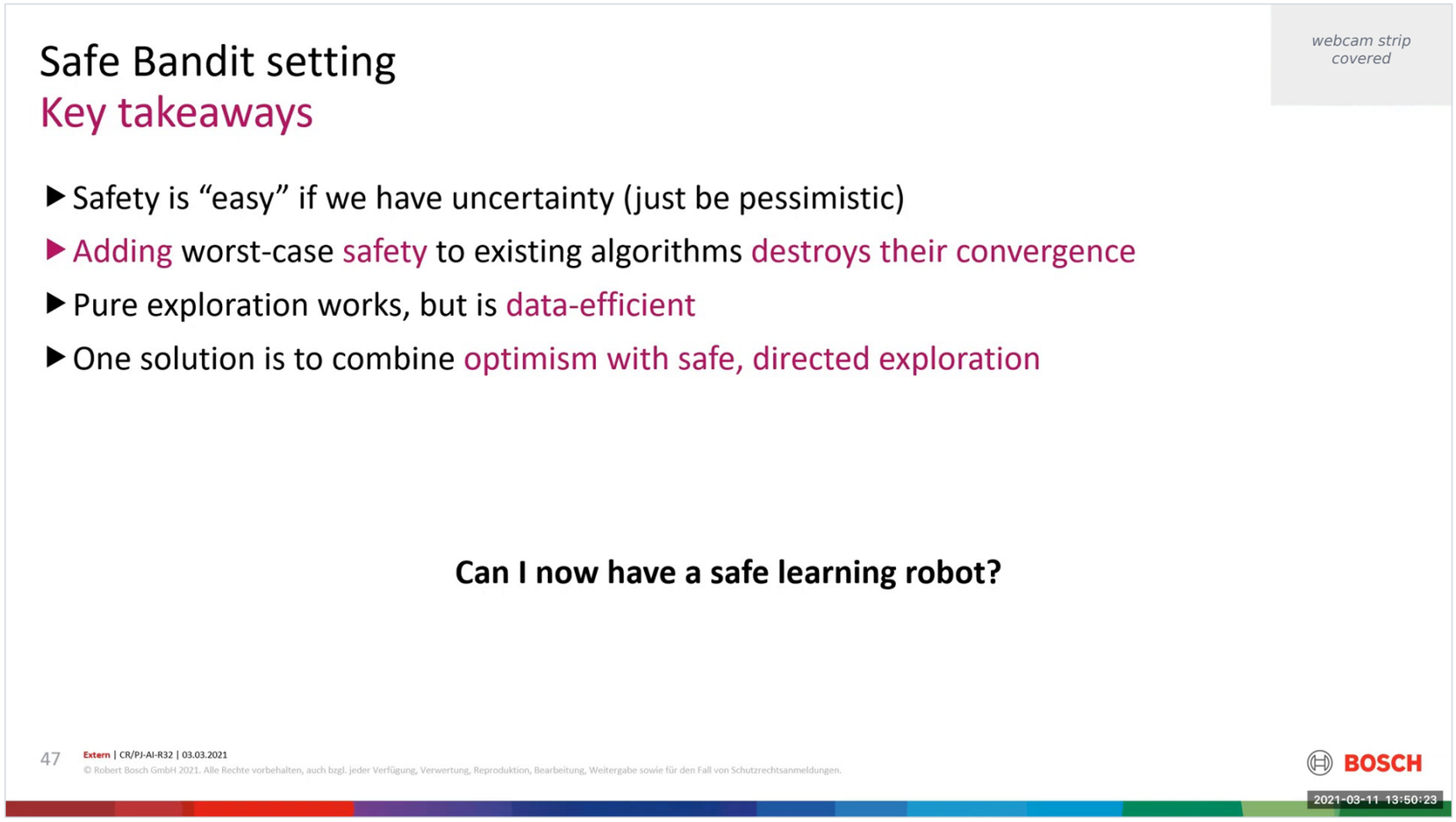


SAFE BANDITS

Video: tuning for different speeds

▶ 1:16:26

- Extension of the same method: tune the drone to fly circles at different target speeds (here 1 m/s and 1.8 m/s).
- Plots show measured trajectories and the parameter space (damping ratio, time constant) as it explores safely.
- From the Q&A: here the safety constraint limited angular velocity, so an onboard camera would not wobble too much.
- Still a bandit-style method: it models score and safety directly from parameters, not the dynamics.

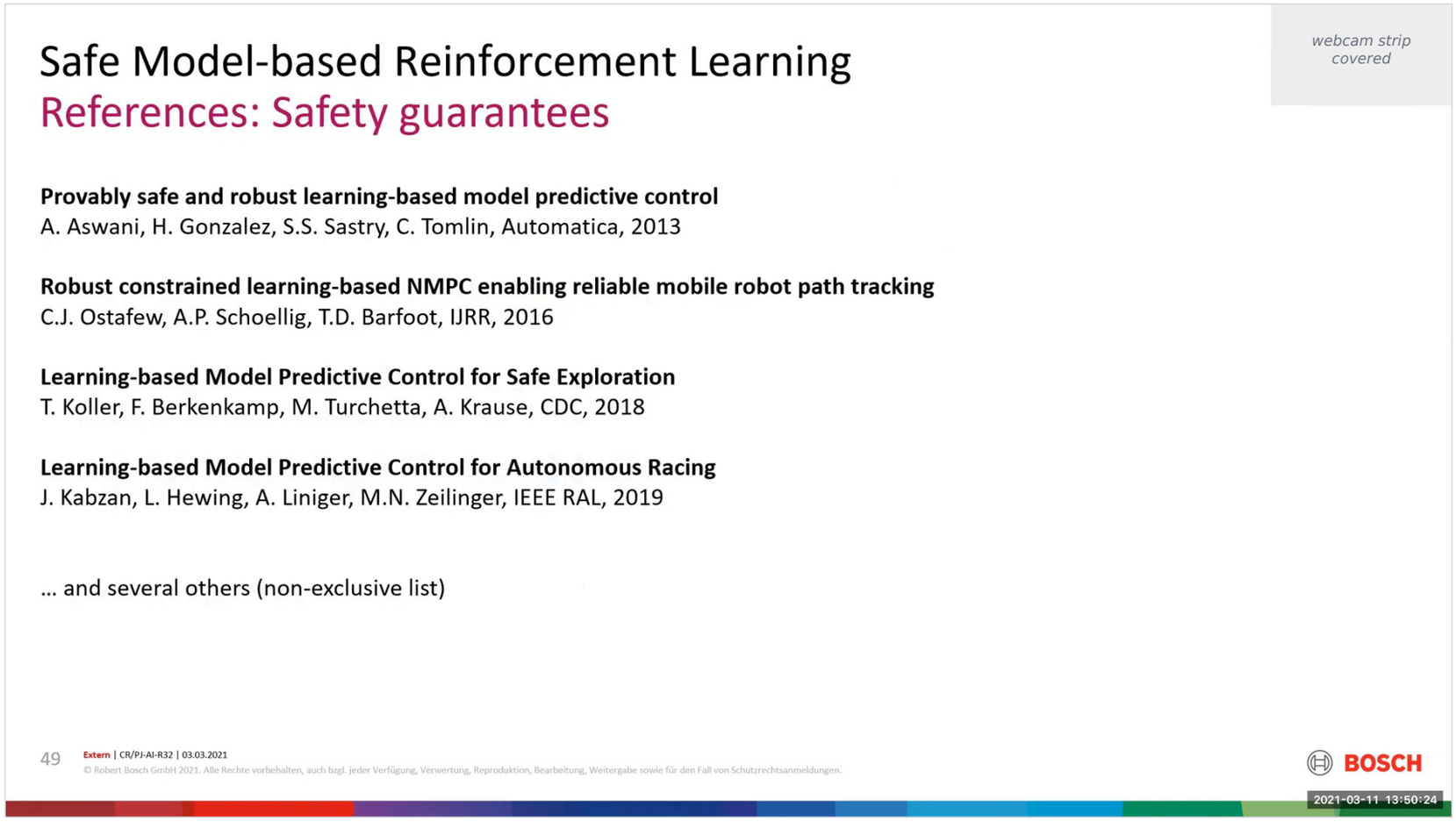


SAFE BANDITS

Safe bandit takeaways

▶ 1:16:58

- Safety is “easy” once you have uncertainty: just be pessimistic. The constrained optimization can still be hard in practice.
- **Adding worst-case safety** to existing algorithms can **destroy their convergence**, even for simple bandits.
- **Pure exploration** works but is not data-efficient.
- One fix: combine **optimism with safe, directed exploration**. “Can I now have a safe learning robot?”



OPEN PROBLEMS AND CONCLUSION

References: safety guarantees (skipped)

▶ 1:18:08

- Shown for under a second at 78:08 while skipping ahead for time. Not discussed.
- Aswani, Gonzalez, Sastry, Tomlin, Automatica 2013: provably safe and robust learning-based MPC (model predictive control).
- Ostafew, Schoellig, Barfoot, IJRR 2016: robust learning-based NMPC for mobile-robot path tracking.
- Koller, Berkenkamp, Turchetta, Krause, CDC 2018: learning-based MPC for safe exploration. Kabzan et al., IEEE RAL 2019: learning-based MPC for autonomous racing. “...and several others.”

Safe Model-based Reinforcement Learning
Safety + exploration

webcam strip covered

Safety and exploration still an open problem in general:

Linear-quadratic (random exploration):

Safely Learning to Control the Constrained Linear Quadratic Regulator
S. Dean, S. Tu, N. Matni, B. Recht, ACC, 2019

Safety combined with uncertainty samping:

$$\pi_t = \max_\pi (1-\rho) J_{reward}(\pi) + \rho J_{\text{information}}(\pi)$$
$$\text{s.t.robust safety},\ \rho \in [0,1]$$

Safe Model-based Reinforcement Learning with Stability Guarantees
F. Berkenkamp, M. Turchetta, A.P. Schoellig, A. Krause, NeurIPS, 2017

Learning-based Model Predictive Control for Safe Exploration and Reinforcement Learning
T. Koller, F. Berkenkamp, M. Turchetta, A. Krause, CDC, 2018

Chance-Constrained Trajectory Optimization for Safe Exploration and Learning of Nonlinear Systems
Y.K. Nakka, A. Liu, G. Shi, A. Anandkumar, Y. Yue, S.J. Chung, IEEE RAL, 2021

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BOSCH

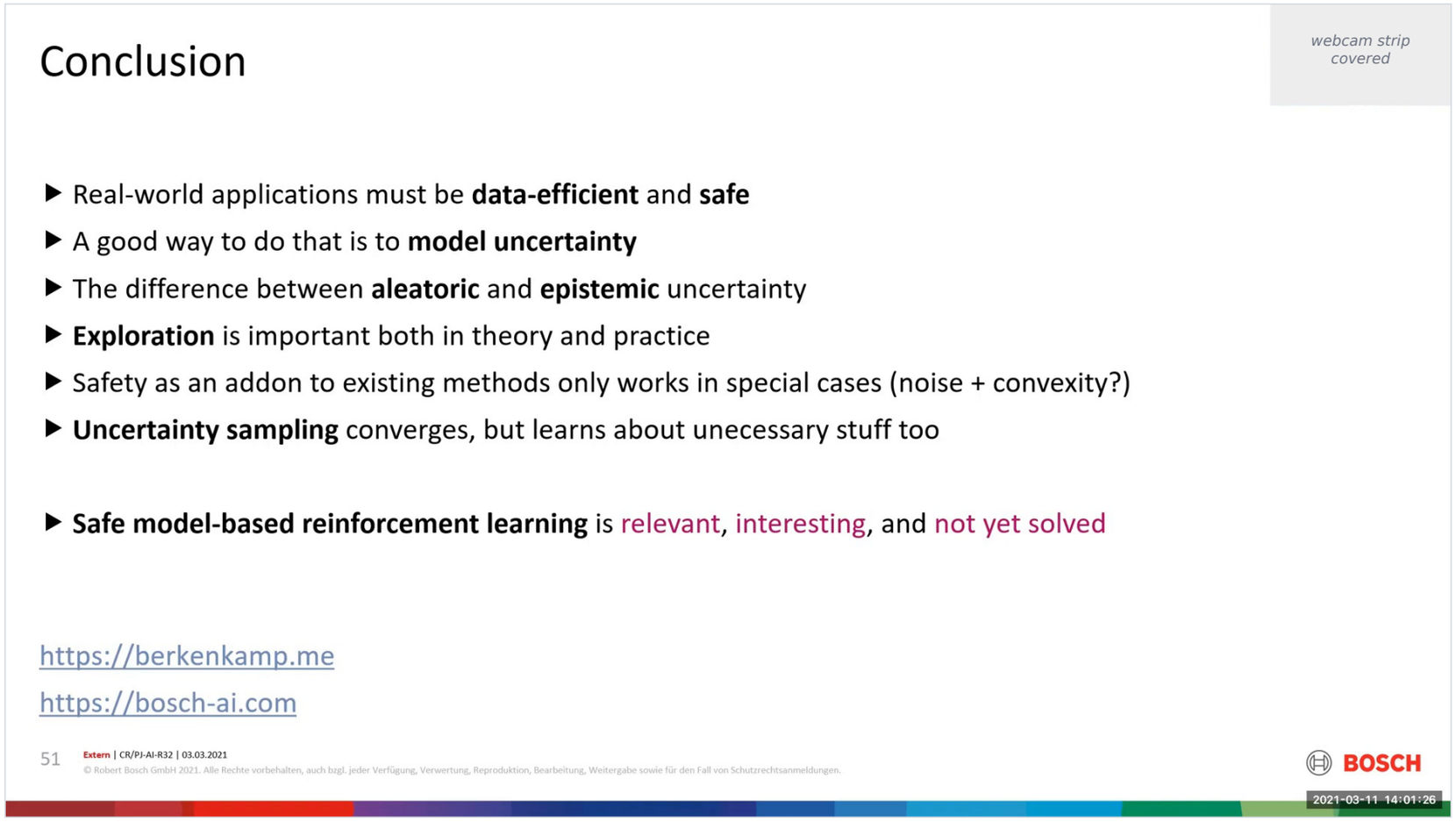
2021-03-11 13:51:18

OPEN PROBLEMS AND CONCLUSION

Safe model-based RL: still open

▶ 1:18:10

- Safety plus exploration is an **open problem** in general.
- **Linear-quadratic** case (linear system, quadratic cost): solved, and random noise is enough to explore (Dean, Tu, Matni, Recht, ACC 2019).
- Otherwise, current methods mix in **uncertainty sampling**: π_t = argmax $(1-\rho)\cdot J_{reward}(\pi)$ + $\rho \cdot J_{information}(\pi)$, with robust safety and $\rho \in [0, 1]$ (Berkenkamp et al., NeurIPS 2017; Koller et al., CDC 2018; Nakka et al., RAL 2021).
- He hopes something better will come along.



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Conclusion and questions

▶ 1:19:04

- Real-world RL must be **data-efficient and safe**; modeling **uncertainty** serves both (where to collect data; pessimism for safety).
- Don’t treat epistemic uncertainty as noise. **Exploration** matters in theory and practice.
- Safety as an add-on only works in special cases. Uncertainty sampling converges but learns things you don’t need.
- Safe model-based RL is relevant, interesting, and **not yet solved**: a good research topic.

Q&A: How was safety measured on the drone? *First video: tracking error mustn’t get too large; circles: limited angular velocity.* ***What about jumps, like knocking a cup off a table?*** *The theory assumes smooth (Lipschitz) functions; MuJoCo smooths contacts; hybrid systems need other tools.* ***Why dislike “explore everywhere”?*** *It works but learns irrelevant things; he expects a cleaner three-way trade-off (objective, safety, optimization), likely found first in simple bandit cases.*